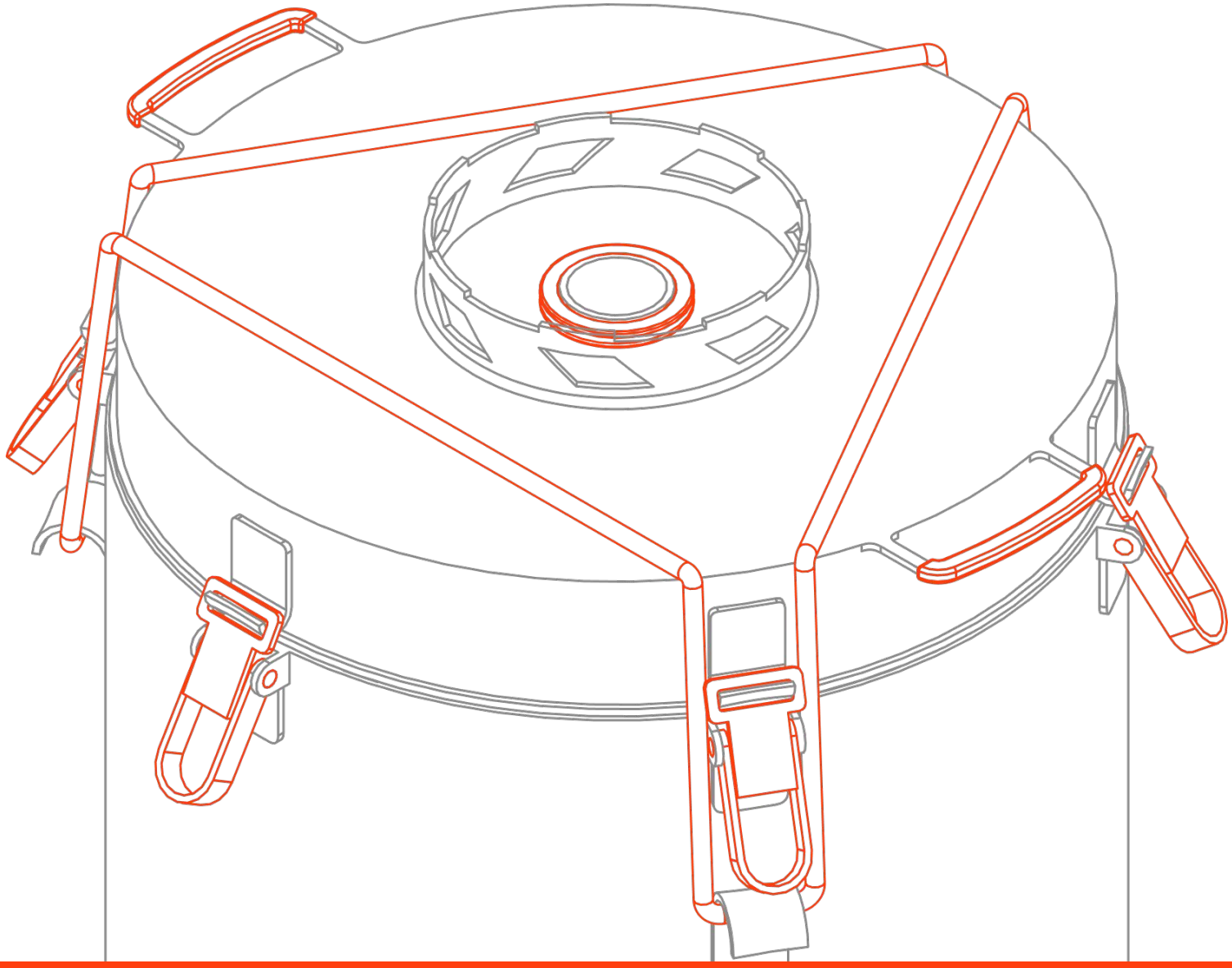

COOKING AND HEATING WITH WET WOOD POST-FLOODING IN RURAL VIETNAM

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i. Abstract

The frequent flooding in rural Vietnam over the typhoon season regularly ruins the local firewood stores that millions rely on, causing severe energy poverty and dangerous conditions for the victims. To provide a localised humanitarian intervention, this thesis presents the context, design, manufacture, testing, and evaluation of a small-scale, natural-draft, monorator wood-gasifier stove designed to dry and burn wet wood after a flood. Given the constraints of the rural Vietnamese context and the delivery of aid within it, the stove was designed and manufactured to use standard stainless steel sheets and insulation, to operate after submersion, and to be used for cooking and heating. While the testing proved the design was only partially successful, it yielded extremely promising results and lessons that, with further testing of a potential redesign, should be fixable, leading to a real solution ready to be integrated into the remote, rural communities most vulnerable to natural disasters.

ii. Acknowledgments

I want to give a huge thanks to my supervisor, Audrey Gaulard, for her help in pushing this project forward and ensuring I was always on track to complete it. I also want to give a huge thanks to GF Custom Fabrications for allowing me to use their workshop and for their help in building their product, and to the owners of Gara Mill House for hosting me during the manufacturing process. I want to thank I. Logan-Rivers and O. Shawn for helping me prepare the risk assessment and safely test the stove. And finally, I want to thank the community on the “Drive on Wood!” forum for their massive help and wealth of knowledge regarding the design and construction of wood gasifiers.

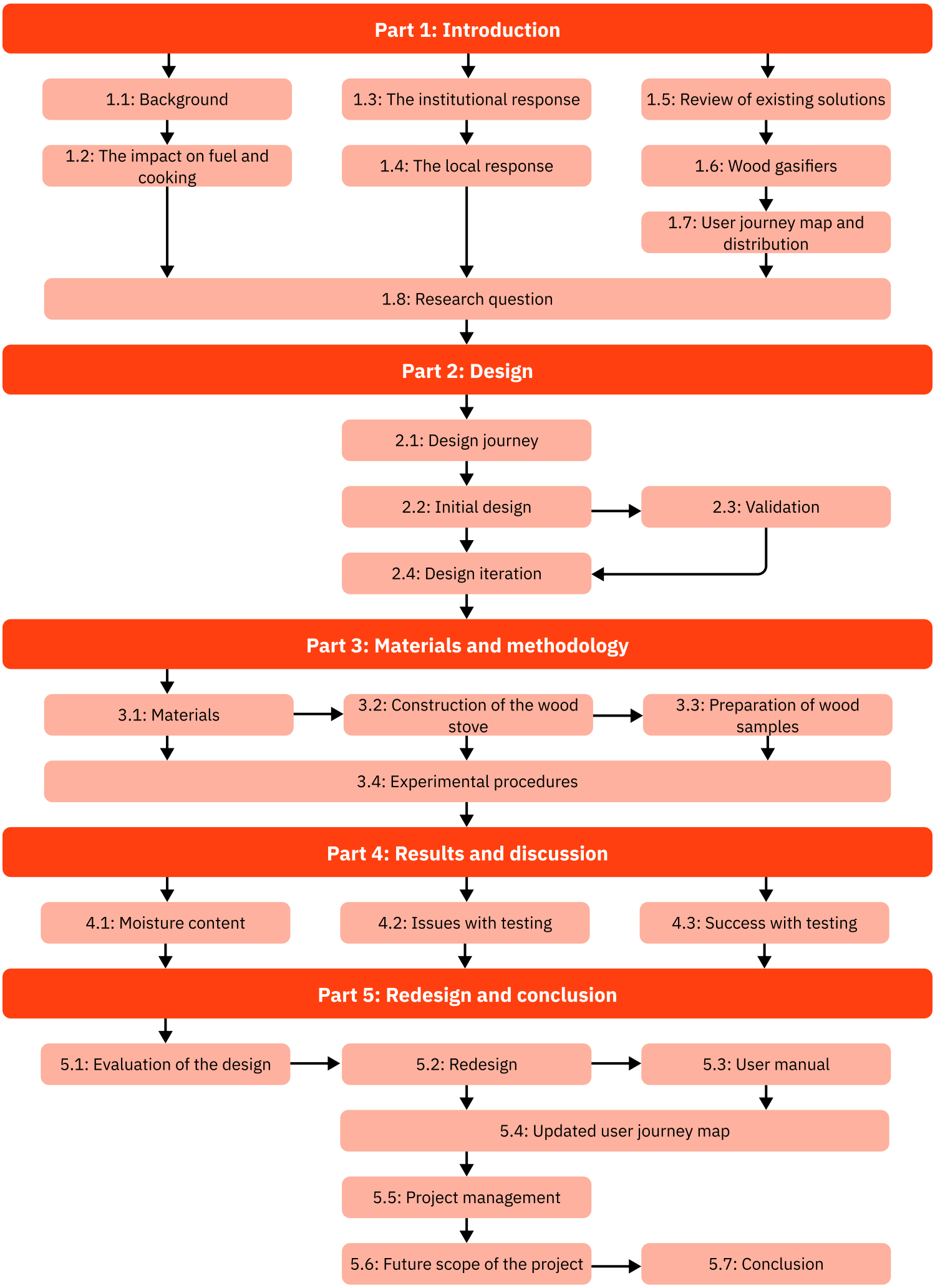
iii. AI Declaration

Generative AI was not used to create or write any parts of this Thesis.

iv. Dictionary of acronyms

Acronym	Definition	Page(s)
CAD	Computer Aided Design	17, 18, 25
CO	Carbon Monoxide	9, 31
COP 30	30th Conference of the Parties (Climate Change Conference)	8, 32
CRA	Climate Resilience Alliance	8, 35
CVA	Cash & Voucher Assistance	13
ESG	Early Stage Gateway	29, 30
ESMAP	Energy Sector Management Assistance Program	9
EU	European Union	4
FFKM	Perfluoroelastomer	26, 27
FTS	Financial Tracking Service	35
GDP	Gross Domestic Product	4
GPA	Global Platform for Action	33, 35
GSO	General Statistics Office	4, 5, 33
HAP	Household Air Pollution	10, 27, 31
HKN	Health Keepers Network	13, 36
ICS	Improved Cookstove	9, 10
IFRC	International Federation of Red Cross and Red Crescent Societies	6, 8, 32, 35
ILF	International Lifeline Fund	13, 36
INFORM	Index for Risk Managment	4, 33
JRP	Joint Response Plan	4, 6, 7, 8
LSG	Late Stage Gateway	29, 30
MAE	Ministry of Agriculture and Environment	37
MC	Moisture Content	21
MEI	Moving Energy Initiative	9, 25, 27
NCA	Neutrosophic Causality Analysis	35
NGOs	Non-Governmental Organisations	7, 8, 13, 30
OCHA	Office for Coordination of Humanitarian Affairs	4, 5, 6, 7, 8, 35
ODV	Open Development Vietnam	4, 32
PAH	Polycyclic Aromatic Hydrocarbons	33
PM	Particulate Matter	5, 9, 10, 31
RA	Risk Assessment	30
SNV	Stichting Nederlandse Vrijwilligers	13, 36
SRV	Socialist Republic of Vietnam	4, 6, 32, 36
UN	United Nations	4, 5, 6, 7, 8, 30
UNDP	United Nations Development Programme	32
UNFCC	United Nations Framework Convention on Climate Change	32
UNICEF	United Nations International Children's Emergency Fund	5, 30, 33, 37
UNVC	United Nations in Viet Nam	5
VDDMA	Viet Nam Disaster and Dyke Management Authority	4, 7, 30, 34, 37
VLE	Village-Level Entrepreneur	13
VMSA	Viet Nam Multi-Sector Assessment	32
WASH	Water, Sanitation and Hygiene	4, 5, 7

v. Structure of the thesis



1. Introduction

1.1 Background

Vietnam is extremely susceptible to climate disasters, ranking 83rd out of 191 on the European Union's (EU) Index for Risk Management (INFORM) (2025). It is most threatened by coastal and river flooding, with Vietnam being rated 9.9/10 on the same Risk Index (European Commission, 2025). With climate change worsening these disasters, increasing the maximum 1-day rainfall by over 40% across the country over the last 61 years (SRV, 2024), and increasing the annual number of typhoons from 5 in the year 2000 (ODV, 2021) to 11 last year (OCHA, 2025a), Vietnam is now reaching a crisis point. Over 2025, the multiple typhoons and floods which have hit Vietnam have wreaked havoc on the population. According to the United Nations' (UN) Joint Response Plan (JRP) (OCHA, 2025a), 3.1M people were affected, 420 people have been reported dead or missing, and over 730 have been injured (OCHA, 2025a). Many impacts affect the victims of flooding, but the 3 key ones I will focus on in this research are damage to households, damage to livelihoods, and damage to Water, Sanitation, and Hygiene (WASH). When combined, they can worsen the long-term humanitarian needs of flood victims, especially around cooking and nutrition.

Damage to households worsens the humanitarian crisis: over 681K homes were flooded, and 67.5K were destroyed or severely damaged, forcing thousands of families to seek temporary shelter with relatives, in schools, and in public buildings (OCHA, 2025a). With water rising 'higher than the windows' (Norkum, 2026), essential household items, such as beds, educational materials, clothing, and cooking equipment, were also completely lost for many families, reducing resilience and lengthening recovery times. Cleaning the mud can often take days, and research by Akter (2021) on the effect of floods on the division of labour in Pakistan found that post-flood, women had to spend on average 2 hours more across all activities, but had to spend 1.7 hours less on care work to make up for the added labour. These added stresses can prolong recovery needs, "particularly among poorer families whose houses were already fragile or located in flood-prone zones" (OCHA, 2025a).

Another factor that affected over 2.5M people was the damage to agriculture and, thus, people's livelihoods: 141K rice fields were flooded, 73,311 cattle were killed, and 5.1 million poultry were swept away over the course of 2025 (OCHA, 2025a). For an industry that is the primary employer for over 92% of people in rural areas (OCHA, 2025b), this has a massive impact on them and their families. In total, the overall economic loss is estimated to exceed Vietnamese đồng (VND) 100T, equivalent to 0.7-0.8% of their GDP (VDDMA, 2025). According to a census run by Vietnam's General Statistics Office in 2020, 14.9% of the GDP comes from the "agriculture, forestry and fishery sector" (GSO, 2020). These economies, though, are at very high risk under the current scenario for Climate Change and sea-level rise in the National Adaptation Plan, where rice production could fall by up to 21.39% by 2100 (SRV, 2024). This not only has immediate impacts on people but, more strikingly, has long-term impacts that can completely remove people's livelihoods and primary food sources, leaving very little time to recover before the next planting season or flood.

Finally, one of the most significant impacts of flooding is the disruption of access to WASH: 230 water systems were damaged, affecting over 2.1M people, and as of December 10th 2025, 216,000 people still lacked access to clean water and safe sanitation (OCHA, 2025a). The damage results from small-scale systems being submerged or destroyed and from solid waste and animal carcasses contaminating surface-water sources (OCHA, 2025a). Without access to piped systems, people become dependent on either bottled water or self-supply systems, which have been rendered unsafe due to contamination from mud, waste, and floodwaters. The destruction of water storage and treatment infrastructure has also left many communities unable to ensure the safety of their

drinking water. Without institutional responses to address these issues, victims will be exposed to polluted water. They will have to deal with skin infections, eye irritation, and gastrointestinal illness, further straining health services.

The combined impacts of household damage, loss of livelihoods, and disrupted WASH access result in a single major impact for victims of flooding: nutrition. The UN's Multi-sector Assessment of the response to Typhoon Yagi highlighted up to a fivefold increase in severe and moderate malnutrition cases in rural villages (United Nations Viet Nam, 2025). With over 107,000 children in need of nutrition support (UNICEF Viet Nam Country Office, 2025) and 5,000 pregnant and lactating women who are at risk of malnutrition (OCHA, 2025b), the scale of the issue is massive. This is not just a short-term effect, according to Assoc. Prof. Dr. Truong Tuyet Mai at the National Institute of Nutrition: 'Many nutrition-related consequences only appear weeks or months later, when people begin to run out of food, clean water, and basic health services' (Tran 2025). Multiple papers have explored the impacts of flooding on nutrition, with Ambikapathi et al. (2021); Gabrysch et al. (2018); Lieber et al. (2022) citing reduced food security and dietary diversity as the main culprits, while the impacts on agricultural production were cited by the Office of UNVC (2025), Lieber et al. (2022), Phuong et al. (2024).

1.2 The impact on fuel and cooking

However, a factor often overlooked is the loss of energy for cooking and water purification, which is directly linked to increased energy poverty post-flooding according to Ogutue et al. (2026) and Okyere et al. (2023). When settlements are remote, transportation and logistics can become complicated, as quoted by Dr Iwona (2022), 'making it difficult to ensure a consistent supply of LPG, ethanol, or improved biomass fuels, such as pellets or briquettes', meaning that people have to rely on firewood, charcoal, and other polluting biomass. Ogutue et al. (2026) and Lee et al. (2021) demonstrate that this is further exacerbated by the existing stock being ruined by the flood, with dry wood stores inundated and the availability of firewood diminished. In areas of repeated flooding, research has shown that up to 94% of households can experience fuelwood scarcity, such as in the Old Brahmaputra downstream zone in Bangladesh (Akther et al., 2010). While mitigation strategies exist, such as using more twigs, leaves, or even cow dung, a similar study in Northern Bangladesh found that 83.8% of women still cite finding fuelwood as a problem post-flooding (Azad et al. 2013). With 18% of the population of Vietnam still relying on biomass (including firewood and agricultural waste) as their primary source of fuel (GSO, 2020), and research conducted in 2019 in the Long An province in Southern Vietnam having found that 70% of people used a surrounded fire with wood as their main fuel (Van Gemert et al., 2019), this is an issue that affects the rural community in Vietnam just as much. Research on the impact of climate change further exacerbates firewood loss and energy poverty in Vietnam (Que et al., 2022; Huong et al., 2019), with the poorest communities in Northern Vietnam, such as Hien Luong in Hà Giang Province, being especially vulnerable, with a firewood vulnerability index of 0.733 (Huong et al., 2019).

The reason this is such a concern is due to what happens when one burns sub-optimal, wet, wood. At 54% humidity, the temperature produced by wood never gets high enough to be usable for cooking, as there is 'no sufficient heat to dry the wood and to heat the food to cooking temperature at the same time' (Gute et al., 2022). Research conducted by Guerrero et al., (2019) comparing 0% and 25% wood moisture content shows a 44% decrease in combustion efficiency, along with an over-10-fold increase in particulate matter emissions (PM2.5). A study on the types of fire-wood typically used by households in central Vietnam shows that the 3 main types are Cratoxylum formosum, Castanopsis kawakamii, and Betula alnoides (Kim et al., 2017). While the type of wood may vary by region, all 3 types and most other selected are hardwoods, indicating a clear

preference for this type as fuel due to its higher energy density (Lunguleasa, Spirchez & Zeeniuc, 2020). This allows me to examine work from Michalec & Niklasova (2006) on other kinds of hardwoods and their water uptake from dry to submerged conditions, which show that after just 72 hours, the moisture content can jump from 12% to around 30%. With floods lasting anywhere from 2 to 10 days before water recedes (Le et al., 2007; Hoa et al. 2008; Hoang, 2025a), and water levels being higher than the windows (See Figure 1 and Figure 2) (Norkum K, 2026; Hoang, 2025a; Hoang, 2025b; Pham Nga, 2025), the existing stockpiles of dry wood are thus likely to be ruined, resulting in households remaining without a reliable fuel source for multiple days or weeks at a time. This is backed by first-hand accounts, where V. Chien, '45, of Hoa Thinh Commune, says his entire hamlet is short of fuel and medicines. All firewood is soaked, making it impossible to cook.' (Nguyen, 2025)

*Figure 1. Flood height having reached above the roof of a rural house (*Hoang, 2025a*)*

*Figure 2. Flood levels over 4m tall on the side of a primary school (*Hoang, 2025b*)*

These impacts are massive on Vietnam's population, but it is the groups that are most vulnerable which suffer the most. It is widely agreed that these groups consist of poor, ethnic minorities, the elderly, women, children and most importantly those with 'climate-dependent incomes' (SRV, 2024; OCHA, 2025a; Ogutu et al., 2026; Lan et al., 2022, retracted 2022). This can be especially true 'for poor households that do not have the resources needed to adapt' (World Bank Group, 2022), where things like the price of fuel sources can be up to twice as much after a flood (Okyere et al., 2023; Nguyen, 2018). This impacts women disproportionately as they are the ones who often must go on long walks to obtain usable firewood (Akter, 2021; Nguyen, 2018), putting them at risk of gender-based violence and increasing both their physical and mental burdens post-flooding (OCHA, 2025a; OCHA, 2025b; Dr Iwona, 2022). They are also at a higher risk of respiratory illness from inhaling pollutants from smoke while cooking (Dr Rosenberg, 2022), with women cooking on average 2.4 hours a day in Vietnam (Van Gemert et al., 2019). As shown above, the effects are exacerbated when wood moisture content is higher, leaving women and children with an even greater impact in the post-flood period.

1.3 The institutional response

The institutional response to flooding in 2025 can be split into 3 branches: the international government response, the Vietnamese government response, and the global NGOs' response. These responses vary in how they are funded, who they target, how they support victims, and their drawbacks.

International government responses are often joined under the UN's Joint Response Plan (JRP) (OCHA, 2025a; OCHA, 2025b) which collects donations from various donors (OCHA, 2025c), often governments or large charities, and then redistributes them across the UN's disaster response branches, various NGOs, and the Vietnamese government to directly help the victims. As of the 17th

of February 2026, the UN's JRP worked with 31 lead agencies and 33 partners (OCHA, 2026). This wide variety of organisations they work with means that they can target almost anyone across Vietnam, with aid having reached 376,362 victims across 21 out of 28 provinces (OCHA, 2026); however, the JRP prioritises the worst-affected provinces first before sending aid to other areas, potentially leaving some victims without relief for longer periods of time. The 3 largest sectors to be targeted by the JRP (OCHA, 2026) were WASH at 171,601 people reached, with a focus on 'providing life-saving water supplies to affected households and evacuation centres' and the 'distribution of critical WASH supplies' such as hygiene kits and water purification tablets; Health came second at 147,000 people reached, with a focus on providing 'in-kind technical assistance' such as vaccines and water purification tablets; and finally Food Security at 100,000 people, with a focus on 'in-kind food assistance to affected households' such as biscuits and food packs. While these large-scale responses should target and help everyone evenly, they come with some major drawbacks: the amount of donations often does not meet how much is required, with only 27.7% of aid having been funded through that plan (OCHA, 2025c); this results in fewer people being reached then originally targeted, with only 376k victims having received aid out of the original 659k targeted (OCHA, 2025a).

Within Vietnam, the government response is structured and managed by the Vietnam Disaster and Dyke Management Authority (VDDMA). The VDDMA organises joint work among the UN, NGOs, and government branches, ranging from national to communal and down to local levels, to facilitate both the prevention and the response to natural disasters (VDDMA, 2026). The government mainly funds the organisation but may also receive donations from citizens, charities, or even the UN's JRP, from which it received WASH and Shelter support in 2025 (OCHA, 2026). At a national level, the VDDMA oversees the distribution of aid around the country, the preparations for emergency shelter and search are rescue, and the tracking of weather events and disasters. At a communal level, they have set up 'Guidelines for Building and Strengthening Rapid Response Teams at the Commune Level' which helps communes to prepare and respond to disasters more effectively (MARD, 2020); they have also produced a similar guideline for the response at a local level (MARD, 2018). Both of those guidelines highlight how local and commune response is crucial in the first stages of disaster: and how they need to complete tasks before a natural disaster (like inspecting and allocating shelters and vehicles, locating and prioritising the most vulnerable people, and preparing infrastructure to deal with the disaster) tasks during a disaster (like monitoring, coordinating support from relevant agencies, and providing direct support to villages, hamlets and neighbouring areas) and tasks post-disaster (like urgently coordinating the rescue and treatment of the injured, providing food, clean water and supplies to those affected, assessing damage and carrying out post-disaster recovery and reconstruction tasks). Unfortunately, though, the VDDMA still faces challenges, especially at the communal and local levels, with issues around information sharing resulting in low coordination and improper funding allocation, leading to some local communities missing critical aid (Lan et al., 2022, retracted 2022).

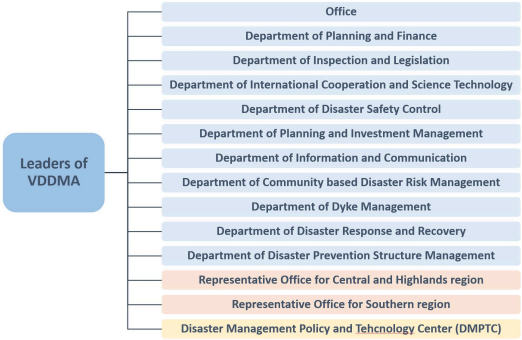


Figure 3. Structure of the VDDMA (VDDMA, 2026)

The third kind of institutional response comes from NGOs and charities. While these again vary in the scale of their operations, they are crucial to ensuring that many people receive the aid they need. Many of these organisations base their targeting on the UN's recommendations, with charities such as Oxfam, Save the Children, and ActionAid supported by the UN's JRP (OCHA, 2026). However, others, such as the Vietnam Red Cross Society (VNRC), receive support from international donations through the IFRC (IFRC, 2024), which can then be redistributed by local teams, offering both physical support and cash and voucher assistance (Norkum, 2026). Other NGOs instead focus on how communities can prepare and be more resilient to natural disasters; for example, Plan International and the Zurich Climate Resistance Alliance have been working with 18 communities since 2021 to improve how they react and prepare to flooding and introduce interventions such as community response teams and Village Saving and Loan groups (CRA, 2026; FRA, n.d.). While this kind of aid is desperately needed by many, there are still some critics of it due to the lack of diversity in the supplies people received, with a victim being quoted as saying 'please, no more instant noodles' (Hoang, 2025) and people asking for cleaning and medical supplies, personal hygiene products and fuel instead (Nguyen, 2025; Phong, 2025).

Unfortunately, all 3 institutional responses share the same major issue: they require that aid be physically delivered to victims. Post-flooding, this can be extremely difficult. With over 1,259 km of road affected in 2025 and a total landslide volume of 14.5M cubic meters (OCHA, 2025a), transporting humanitarian aid by truck and other large vehicles can become impossible, limiting how far and how quickly aid can be delivered to victims. Research by Xu W et al. (2021) and Zarei, Carrasco-Gallego & Ronchi (2019) on the resilience of humanitarian aid supply chains to flooding highlights that transportation capacity is critical to supply chain sustainability and that improving resilience can enhance last-mile delivery capabilities, ensuring rapid access to aid in affected areas (Nguyen, 2025). With institutional support often prioritising economic aspects in rural areas as a measure of success, a push towards integrating a more local community response and knowledge has been advocated (Bruun & Olwig, 2015).

1.4 The local response

As Lucas Turmera (COP30 | German Climate Pavilion, 2025, 10:00) put it at the COP30 Bélem panel: 'In the absence of institutional response, community-led actions were often the first or only response which occurred'. Across Vietnam, there are many stories of communities coming together to support one another in the absence of an institutional response. The news has been flooded with stories of compassionate souls who set up makeshift kitchens to help those in need (Duong, 2025; Thao, 2025; Nghia, 2025; Minh, 2025). For some, this is a massive impact, Duong P. (2025) from VNExpress reported that 'their meals go deep into hamlets, to places that residents say "no other group has managed to get into yet"'; but the most evocative quote was from when they managed to get warm food to an elderly: 'When they handed him a hot box of rice, he burst into tears like a child. "I haven't eaten for days," he said. "I'm so hungry"' (Duong, 2025). The local response isn't limited to just makeshift kitchens: neighbours prepare and, if needed, borrow essentials from each other, including rice, firewood and dry food (Bao, 2025; Nguyen & Võ, 2023); neighbours with sturdy, multi-story houses shelter any vulnerable person who asks for aid, in one case, hosting 50 people in rooms 'filled with the sounds of children crying from hunger and elders groaning from the cold' (Pham, 2025; Phong, 2025); and communities also gather along the 'riverbank to collect firewood, despite the strong currents of the floodwaters and the potential dangers' (Nhu, 2025). Meanwhile, people flock to social media to request volunteers and aid (Erika, 2025), volunteers set up search-and-rescue operations to aid the most vulnerable (Minh, 2025), and companies that would normally use boats and jet skis for tourism shift to using them to transport people and supplies (Thao, 2025).

While the internet is flooded with heartwarming stories of neighbours helping neighbours, the need for this often stems from an inadequate institutional response. When aid doesn't arrive in time, or the aid delivered doesn't meet the current needs of the victims, communities are left to their own devices to try to survive and repair the damage caused. This can result in horrendous conditions for victims, where they are found cold and shivering, in situations where 'for the first three days they could not cook or light a fire' (Phong, 2025), only able to sit there waiting for the wrong kind of aid to arrive. This is why remote rural communities must be independently resilient to flooding and disasters, especially with respect to firewood and fuel. While they are being hounded by floodwater, wet clothes, disease, and a lack of food, having the comfort of being warm, eating a cooked meal, and drinking sanitised water without having to wade miles to collect dry firewood should not be a fantasy but a reality.

1.5 Review of existing solutions

The Moving Energy Initiative (MEI) produced a paper reviewing cooking systems in humanitarian settings 'to offer guidance on the design and implementation of improved cooking systems in displacement situations' (Vianello, 2016). This serves as an excellent toolkit for reviewing existing stoves, their issues, and considerations for my design. The paper applies the ESMAP typology, produced by the World Bank (Putti et al., 2015), to classify improved cookstoves (ICS) across 5 different categories:

1. Legacy and Basic ICS: Small functional improvements in efficiency over baseline technologies, typically artisan manufacture.
2. Intermediate ICS: Rocket designs with significantly improved efficiency, some manufactured with high-end materials.
3. Advances ICS: Fan-jet or natural-draft gasifiers with high combustion efficiency; may require pellets or briquettes.
4. Modern fuel stoves: Use fossil fuels or electricity; high fuel efficiency; very low CO and PM emissions.
5. Renewable fuel stoves: Energy from renewable sources.

Each stove type requires specific fuel types, which have varying running costs, affecting desirability for new adopters. Legacy, Basic and Intermediate ICS, and Renewable have the lowest annual running costs per person, with means of $24, $21, and $26, respectively (Vianello, 2016); this is due to their ability to use cheaper fuels like wood and charcoal and to have lower maintenance costs. Advanced ICS are slightly more expensive at $30, but modern fuel stoves are the most expensive, with a mean of $219 (Vianello, 2016). However, this financial analysis must be examined with scrutiny, as fuel, maintenance, and implementation costs are highly context-specific, especially in the aftermath of flooding.

The paper also produced an extensive list of design considerations, highlighting the need to consider not just cost and efficiency but also cultural factors that affect the stove's use. This brings us to the following design considerations, which will be used in the design process later on:

1. Durability: Withstand heat, use and damage.
2. Stability: To ensure safety when in use.
3. Ease of lighting: For convenience and speed.
4. Cooking speed: Very important when water is routinely boiled for staple dishes.
5. Fuel type: Needs to be able to use fuels which are available, affordable and sustainable.
6. Portability: Where outdoor cooking or transport is required.
7. Radiant heat output: If the stove is used for heating as well.
8. Fit with existing utensils: No new pots are required.

9. Visual attractiveness and build quality: To make the stove an appealing investment.
10. Price and availability: Weighted against existing products.
11. Clean burning performance: Reduce Household Air Pollution (HAP).

However, there are some factors which the paper did not consider, as it did not place the stoves in the context of post-flooding or post-natural disasters, but only in the use in refugee camps. The most critical missing requirement was that most of the Legacy, Basic, and Intermediate ICS use "optimal wood" (below 20% humidity, ideally below 15%) or an alternative biomass, which is more expensive to acquire and transport. Two examples of this are the Frontier Stove (Anevay Stoves, n.d.a), which recommends the use of their 'Eco Fuel' (5% humidity) and cite a worry about burning 'wet or resinous' wood as their flute can get blocked (Anevay Stoves, n.d.b); or the SilverFire which was a purpose made stove for disaster relief in Puerto Rico and Myanmar to use available biomass for smoke free cooking, however, it also suffers from the same issue of requiring "dry optimal biomass" to work (SilverFire, n.d.).

The second issue the product needs to consider is water resistance and its ability to continue working even after having been submerged. Some stoves, specifically Advanced ICS or Modern fuel stoves that use LPG, struggle with this, as thick silty water clogs the thin gas pipes, and fuel is difficult to transport between affected zones due to road damage that blocks trucks and cars. Even bio-solutions, which do not require the transport of fuel, such as the GreenGen Biodigester (GreenGen, n.d.), struggle with flooding as they are very susceptible to damage from the water, but even more importantly, aren't portable enough to be brought up to higher levels and used by people stuck on top of inundated houses.

1.6 Wood gasifiers

With a focus on the ability to burn wet wood, my research brought me into the extensive world of wood gasifiers. Wood gasifiers work by combining the thermochemical processes and reactions of drying, pyrolysis, oxidation and reduction to turn solid fuel into a combustible gas (Basu, 2010). This is done by partially combusting fuel in a closed container, with a small amount of air delivered to raise temperatures high enough for cracking to occur (Rabby et al., 2023). The appeal of these methods for burning wood is their much higher thermal performance compared to traditional cookstoves, ranging from 5.88% to 91% depending on the design (Rabby et al., 2023), and lower PM2.5 emissions when tested against improved cookstoves (Carter et al., 2014). Both factors help reduce the need to collect firewood, improve the ability to burn wet wood and reduce the dangers of HAP to users.

The design of a wood gasifier stove is crucial to its success, especially with fuel limitations. The type of gasifier and its design can vary massively. Still, across all designs, some categorisation can help clarify how they work (The following categories are the ones most relevant to my wood gasifier stove and do not consider more complex processes for industrial use cases, such as fluidised beds). The first and most generalised way to categorise wood gasifiers is between updraft and downdraft gasifiers: in updraft, the gases flow from the bottom of the container through the different reaction zones upwards and are then mixed with a secondary flow of air and burnt; meanwhile, in downdraft, the produced gas flows downwards through the reaction zones and is extracted from the bottom of the gasifier. The second distinction one can make is between natural- and forced-draft gasifiers: the difference is whether the airflow is supplied by an external source, like a fan, or naturally by the gasifier's design. Finally, one can differentiate between top-lit and bottom-lit gasifiers: top-lit burns the gas above the reaction stages, while bottom-lit extracts the gas and burns it externally. Figure 4 and Figure 5 show the categorisations explained above, and the different reaction zones.

Figure 4. Top-lit Natural Updraft Gasifier

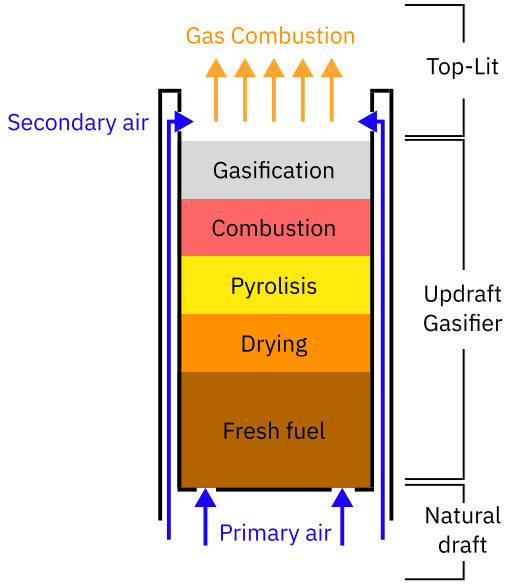
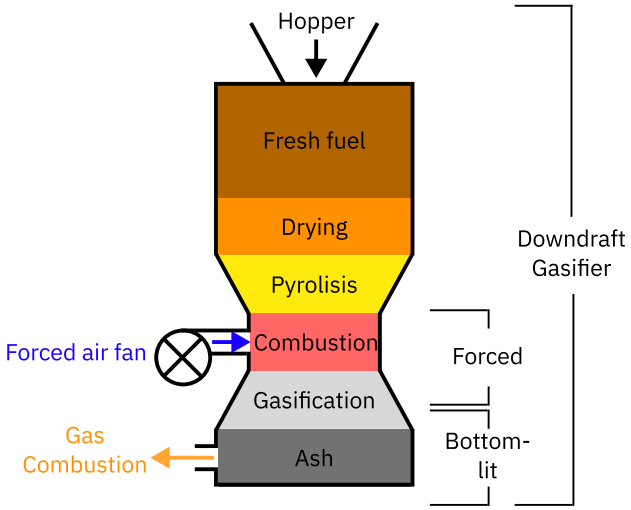
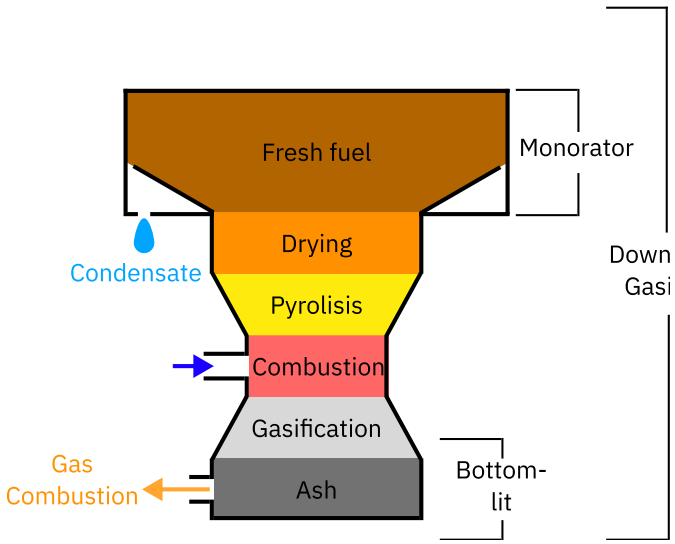


Figure 5. Bottom-lit Forced Downdraft Gasifier



Each of the key categorisations has its own advantages and disadvantages. Updraft gasifiers can handle much higher moisture content, up to 60%, compared to downdraft's 25% (Basu, 2010); however, they struggle with issues related to fuel reloading if more cooking is required, up to 50x higher tar production, and a lower gas exit temperature (Basu, 2010). Between forced and natural draft, the advantage of forced draft is a massive increase of efficiency with "the potential to greatly reduce fuel consumption and minimize indoor air pollution" (Rabby et al., 2023); however, the requirement for an external air pump could lead to the inability to use the stove if no external energy source is available, such as if electricity is cut off post-flooding. Lastly, the difference between top-lit and bottom-lit is minimal in terms of performance, but top-lit can often appear easier to use and offer more flexibility for users.

Figure 6. Bottom-lit Monorator Downdraft Gasifier



The final key source of information I used to understand better the world of gasifiers was the "Drive On Wood!" forum. This forum contained the knowledge of many people who had built, operated, and experimented with different gasifiers over the years. The first discussion I joined was specifically about the use of wet wood as fuel (Alex, 2025). The consensus was that this would not work if the purpose of the gasifier were to run internal combustion engines, as too much energy would be wasted drying the wood; however, for internal heating or for heating water, the system could still work. Another discussion around problems with using wet wood to run a gasifier highlighted the possibility of using exhaust fumes, heat from external piping, and heat from the engine to dry the wood before it enters the hopper (Veljko, 2024). A recurring recommendation between posts was to use a "monorator" above a downdraft gasifier to dry the wood (See Figure 6). Compared to a

traditional downdraft gasifier, where the tall, lean container for the wood has chaotic gas circulation, a monorator has a low, wide hopper to increase vapour condensation on the colder sides of the container and enhance convection currents within the design (Pr. Kyrklund, 1945). Lab testing of a monorator against a standard gasifier by Professor Kyrklund (1945) showed that while there was little improvement in performance between oven-dried wood at 8.6% moisture and air-dried wood at 19.5% moisture, the monorator gasifier maintained performance with damp wood at about 47% moisture. In contrast, the standard gasifier failed in less than an hour (Pr. Kyrklund, 1945).

1.7 User journey map and distribution

The user journey map explains the key stages victims go through during a flood and is based on the information collected above (See Figure 7). The current user journey map does not have a potential intervention in it, but has been updated in section 5.4 to show where the intervention could be placed.

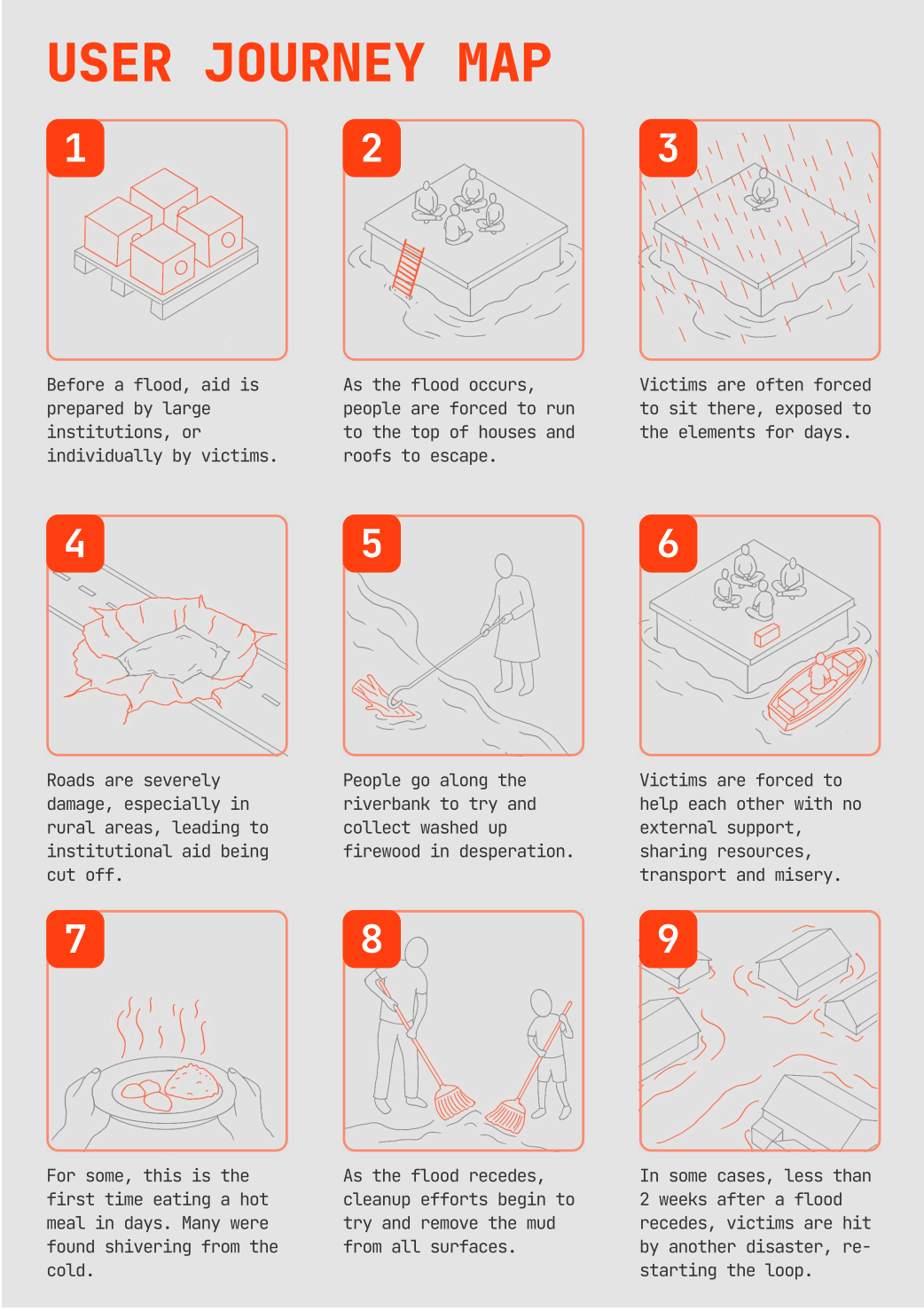


Figure 7. User journey map

Depending on which distribution model I use, my intervention could be delivered to victims in stages 1, 6, or 8. Research conducted by SNV on 'Successful Cookstove Distribution Models' (2015) highlighted three distinct models employed by non-profit institutions:

1. Village-level entrepreneur (VLE): Stoves are distributed at the local level through micro-distributors, leveraging entrepreneurs' knowledge and selling door-to-door. This results in low investment costs and scalability, but can pose issues if the product is unknown and there's limited control over where and to whom the product is sold.Piggybacking: Suppliers work with other delivery partners, such as wholesalers, supermarkets, or hardware stores, to reduce the costs of "last-mile" distribution. The advantages include lower investment costs, no additional infrastructure needed and shorter time to establish markets. However, there are severe issues with markets quickly becoming oversaturated, limited control and engagement from network actors.
2. Proprietary sales network: An entire distribution network is set up, ranging from manufacture to DTC sales. This allows for full control, avoids middlemen markups, and enables more financing for victims. However, it is also very costly and may be difficult to reach customers through it.

The study, however, also recommends combining aspects from all 3 models to improve the sales and last-mile distribution of the device. In rural Vietnam, marketing should be reinforced by respected figures within the community, such as health workers or teachers, and focus on immediate personal benefits such as saved money or time' (Vianello, 2016) by avoiding the need to collect dry firewood. A combination of VLE and a Proprietary sales network would enable victims in remote rural villages to be reached and to exploit Cash & Voucher Assistance (CVA) received by the communities, facilitating easier financing for victims.

Successful examples of these distribution models from non-profit organisations include the Health Keepers Network (HKN), which uses a 'community-based distribution channel' made up primarily of women VLEs to sell cook-stoves door-to-door (HKN, 2026). Their success comes from workers being trained in sales, business management, and health promotion, and from their own experience with the product, which allows them to give trusted advice to their customers. Another good example that leverages Cash & Voucher assistance and donations to supply stoves at a highly subsidised price is the International Lifeline Fund, which uses the piggyback model and partners with supermarkets, NGOs and other aid groups to deliver the stoves at a subsidised price (ILF, 2026).

1.8 Research question

The proposed design below aims to answer the question: Can a small-scale wood gasifier stove be used for cooking and heating with wet wood post-flood in Vietnam?

As such, the evaluation of the design will need to account for a few important factors:

1. The success will be evaluated by analysing the ability of the wood gasifier stove to boil water, sterilise it, and cook rice in it.
2. The maximum moisture content of wood that the wood gasifier stove can handle.
3. Portability of the stove so it can be transported to affected victims when roads are too damaged for cars, vans or other traditional large-scale modes of distribution.
4. The stove's ability to warm the surrounding area to heat victims.
5. How well the design meets the Moving Energy Initiative's 11 design considerations (Vianello, 2016).

2. Design

2.1 Design journey

The design journey began by constraining the design to fit the user's and contextual needs. The easiest constraint to create was dimensions and weight. In terms of dimensions, the maximum size that the outside of the stove was constrained by, looking at the available transport which could carry it after a flood, by looking at the back of a standard moped and the space available on a small dinghy, the maximum size the device could be when packaged is a 450x450x450mm cube. In terms of weight, I first looked at what the maximum loading capacity for the most popular motorcycle in Vietnam is, as an estimate, that being the Honda Vision 110 as of 2022 (Motorbikes Dung, 2022), which has a carrying capacity of 165kg (Honda, 2026). Assuming a passenger weight of 80kg, that gives us a maximum carrying weight of 80kg. However, the stove should be portable enough to be carried by a single person, so I set an unloaded limit of 20kg to ensure a person could carry it up stairs or to a roof in case of emergencies.

When it comes to flooding, the first constraints I needed to look at are the device's ability to handle being submerged, meaning it should not have any small perforations that can get clogged by mud or silt in floodwater, and should have no areas where water can pool. The device should also be made from materials that are unaffected by floodwater, so the material choice should focus on stainless steel, ceramics, and other materials that will not be damaged if submerged for extended periods. Finally, the stove should be able to handle drying and burning high-moisture wood. This brings us to the monorator design for the gasifier stove, which should be included to help with drying the wood. Another important factor in the gasifier design is that it should operate on natural draft, meaning no external energy is required to pump air into the system to run the gasifier. For safety and efficiency, a layer of insulation should also be placed around the gasifier to reduce radiant heat. Finally, while researching the design of a wood gasifier, I was unable to find information on the required dimensions, volume of wood or airflow needed to build one successfully; however, I was able to find a paper regarding the target bed height, which stated that the height of the gasifier chamber should be around 0.5 times the diameter (Al-Farraji, 2017). The final constraints I gave to the design were based on the design considerations from the Moving Energy Initiative (Vianello, 2016), with a focus on the ability for the design to work with existing utensils, so no new pots are required, the stability of the stove to increase safety, and the durability for it to be able to withstand heat and damage.

2.2 Initial design

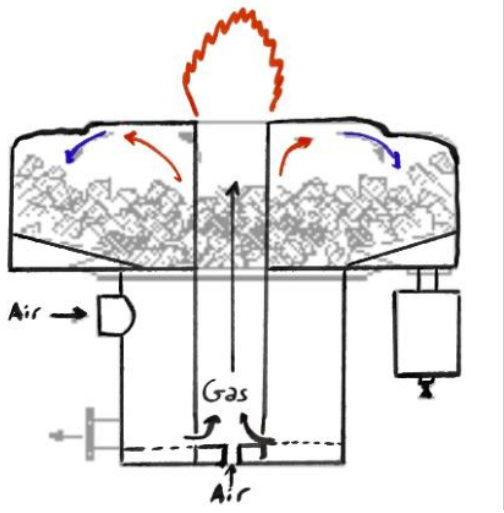


Figure 8. First sketched design

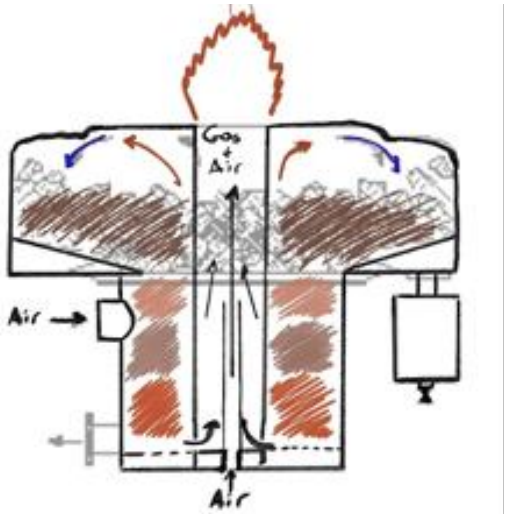


Figure 9. Iterated design for forum

Initial design work focused on creating an improved version of Professor Kyrklund's monorator (Pr. Kyrklund, 1945) and then sought feedback from the “Drive On Wood!” forum for validation and potential improvements. The first idea I had was to funnel the hot syngas out of the device and instead through the centre of the stove to heat the wood in the top funnel, creating stronger convection currents, which would allow for a more efficient design (See Figure 8). The other minor design change I added to it before asking for validation is to add a secondary air channel passing through the middle of the inner pipe, which would mix fresh air with the syngas for a cleaner burn, while using the chimney effect from the rapidly rising hot air to pull fresh air into the system (See figure 9).

2.3 Validation

Contacting experts on the “Drive On Wood!” forum proved crucial to my design work. In the post I raised (Rastelli, 2026), the key questions I wanted answered was around if tar build up mattered as much since I wasn’t running an engine, the usability of a monorator design, whether the syngas fumes need to be cooled down before being burnt or if that’s only a requirement for driving an engine, and finally feedback as a sanity check on my early design.

The first important bit of feedback I received concerned the energy required to dry wood (Šuštarič, 2026): Assuming wood at 40% moisture content, so 40% of the weight is water, it will require a large amount of energy to be lost to the system to evaporate all that water. A rough estimate from that user indicates that around 10% of the wood’s weight in energy is used for evaporation, meaning that more air is needed for the drying, pyrolysis, and oxidation stages. The second key feedback I received from Mackey (2026) was to use the bottom pipe as the air intake to create a better draft at that point, as air rises through the centre pipe, allowing me to place a valve there to control the incoming air and, in turn, the strength of the gasifier. A minor comment also made here from Mackey (2026) was to deepen the bottom ash catcher to allow for more ash to be collected.

Finally, the last key set of recommendations I received came from a user, Southerland B (2026), who had worked with poor farmers in East Africa, focusing on efficient stoves that used sticks as their primary fuel. The first point he raised was that using ‘previously dried biomass that has become water-soaked would be my preferred fuel over fresh cut green wood’ (Southerland, 2026), due to it producing less tar due to a lack of sap in the wood. He also recommended combining a charcoal kiln with a resort to burn wood efficiently. His system would work similarly to my design, where the ‘upward flowing heat from the starter fire [initialises] pyrolysis and then becomes downdraft gas jets when the 30-gallon lid is put on top’ (Southerland, 2019). However, his system required a forced draft to run, which was built from adapted hand blowers, a potential solution for my design. The success of his design and its similarity to mine are promising for mine.

Some criticism around my design raised by users was around the use of a wide hopper above, as it could cause wood to be stuck in the wide edges (Šuštarič, 2026); however, this is based on a not very well-known design for the monorator gasifier, and has an angled base to funnel wood into the bottom section, so that risk is easily mitigated. Another recommendation was to create large piles of wood and burn them to produce charcoal (Unruh, 2026). This is not an acceptable solution for me, though, as it requires a large open space, and if a user were stuck on a roof or in another confined space, it would be impossible for them to set it up.

2.4 Design iteration

Based on the feedback I received from the forum, I decided to estimate the amount of wood required to cook 5 portions of rice (375g), accounting for wood moisture, as an input for the device's

internal dimensions. To start, the amount of wood can be worked out by looking at the amount of energy needed to raise 750ml of water from 24° to 100°C and maintain that temperature for 23 minutes (which is the time to boil water and cook rice), while accounting for heat loss from evaporation, the steel pot being heated up, and inefficiency within the stove.

To heat 750ml of water to 100C, it would require the following amount of energy (where Q = energy, m = mass of water, c = specific heat capacity of water, ΔT = change in temperature):

$$Q = m * c * \Delta T = 0.75 \times 4184 \times (100 - 24) = 238,488 J$$

The energy loss from evaporation with a lid on can be estimated at around 5% per hour, or about 37.5ml of water. The energy loss over 25 minutes is thus (where Q = energy, m = mass of water, t = time, H = heat of vaporisation):

$$Q = m * t * H = 37.5 \times \left(\frac{23}{60}\right) \times 2260 = 32,488\ J$$

We also need to account for the heat absorbed by 375g of rice, which has a specific heat capacity of around 1800 J/kg · °C-1 when uncooked (Iguaz et al., 2003). Using the same equations as before (where Q = energy, m = mass of water, c = specific heat capacity of water, ΔT = temperature change):

$$Q = m * c * \Delta T = 0.375 \times 1800 \times (100 - 24) = 51,300 J$$

Finally, the heat absorbed from a standard 2L pot, or a weight of around 0.82kg (86), at a specific heat capacity of 502 J/kgC^{-1} (Engineers Edge, n.d.) gives us an energy requirement of (where Q = energy, m = mass of water, c = specific heat capacity of water, ΔT = temperature change):

$$Q = m * c * \Delta T = 0.82 \times 502 \times (100 - 24) = 31,285 J$$

The total energy required at 100% efficiency is thus:

$$Q = 238,488 + 32,488 + 51,300 + 31,285 = 353,561 J$$

To calculate how much wood is required to produce that energy, we need to look at how much energy is released from the heat of the combustion of wood char, the efficiency of the stove, and how much energy is lost to the evaporation of water. The heat of combustion of wood char produces between 30.8 and 34.3 kJ/g (MacLeod, Law & Hadden, 2023). A modern wooden stove has an efficiency of around 35% when using wet wood (Gute et al., 2022), and around 10% of the wood's energy must be used to evaporate the water (Šuštarič, 2026). This means a minimum mass of wood of:

$$M = (353,561\ \div 34,300) \div 0.35 \div 0.1 = 294.5\ g$$

Assuming a hardwood density of 0.947 g/cm3 (XHOME Sai Gon Marketing Team, 2023), the volume required of wood undergoing char will have to be at least:

$$V = 294.5 * 0.947 = \ 278.9\ cm^3$$

Assuming a gasification zone height of 3.5cm and accounting for the size of the inner tube, this allowed me to set a minimum inner-diameter dimension for the gasifier, of at least 5cm. Together with the recommendations from the “Drive on Wood Forum”, I decided to add a simple funnel to the top of the inner gas pipe to act as a Venturi ejector, allowing me to speed up the hot air more, and so increase how much fresh air was being pulled through the secondary air pipe. Primary air intake pipes were also added to feed air further up the gasifier from below the design, allowing all 4 air intakes to be controlled by a single valve. Figure 10 shows initial measurements with improvements for the design.

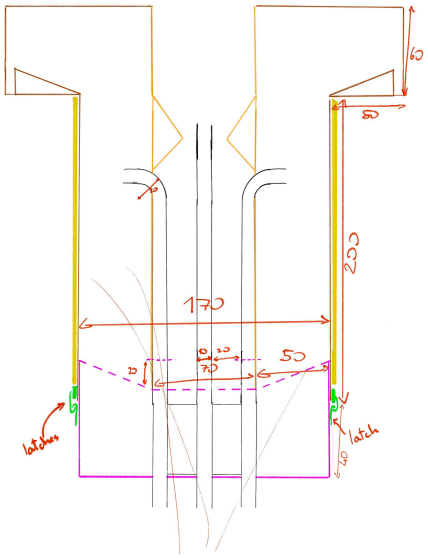


Figure 10. Initial measurements

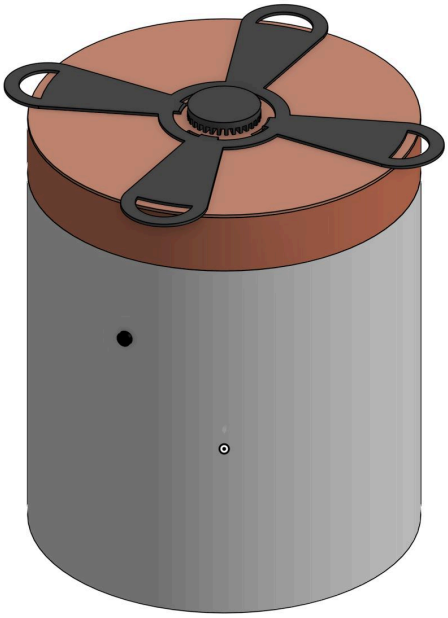


Figure 11. V1 CAD design

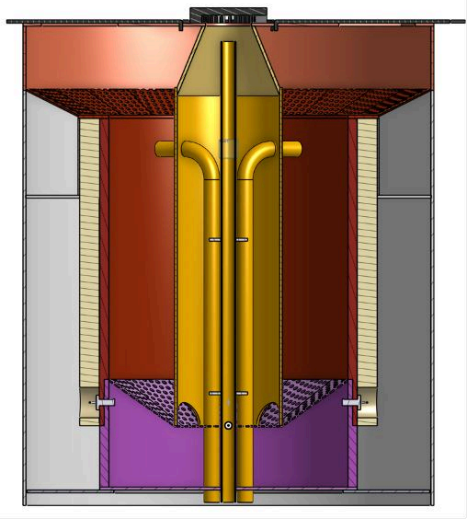


Figure 12. V1 cross-section

From this design, I began work on a V1 CAD model, as shown in Figures 11 and 12. The initial design replaced the latches above with simple pins, as they could more easily bypass the insulation, and included a rotating lid system that allowed a user to twist the lid to insert the wood. The design is made up of 5 key components, which can be separated from each other for cleaning purposes:

1. The centre yellow section is the gas system, both bringing fresh air into the system and exhausting the hot syngas from the top, which is drawn through the large holes above the purple grate.
2. The purple section at the bottom is the ash catcher, which collects all the ash produced by the wood above for later disposal.
3. The red sections consist of the monorator hopper and insulation, where most of the gasification stages will occur.
4. The grey case protects the user from high temperatures, protects the inner sections from damage, and collects water in the channel shown above, which can be drained through the black plug shown in Figure 11.
5. The black section consists of the lid, which can be opened and closed by twisting it, the black plug for draining water, and the diffuser at the top of the stove to spread the gas and cook on top of it.

Some issues I found with this design were related to operation: the twisting system for opening and closing the lid would not provide a sufficient seal and could potentially injure the user if it gets too hot. The pins that held the red and purple pieces together were also suboptimal because they were difficult for the user to reach. Finally, the water channels did not need to be that high, as the system could drain out of the bottom of the casing. The last issues to be addressed by V2 of the CAD model include the lack of a stand, the lack of control over airflow entering the system, and some worries around the manufacturing of the device, with the overlaps between the red and purple sections, originally meant to seal, being too difficult to produce.

Figure 13 shows the setup of the device for the V2, Figure 14 the cross-section view, and Figure 15 the device when it's packaged up for transport, with the stand acting as a protective case for the valve below, and a utility cord attached to the feet to hold the whole product together.

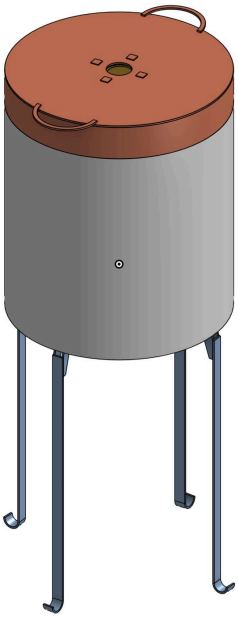


Figure 13. V2 CAD design

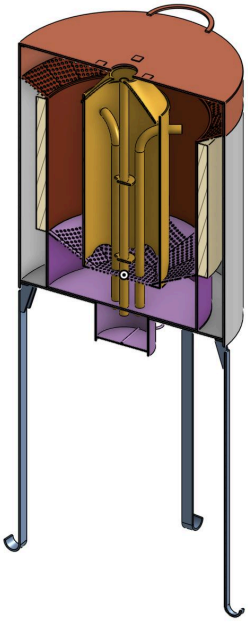


Figure 14. V2 cross-section view

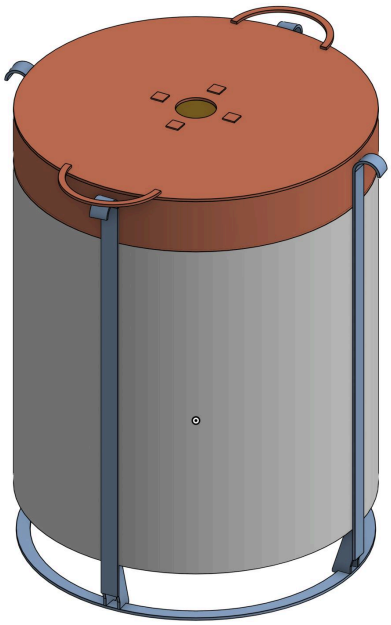


Figure 15. V2 packaged design

3. Materials and methodology

3.1 Materials

The materials for the manufacture of the small-scale wood gasifier are as follows: 307 stainless steel sheet, mild steel sheet, 10mm mild steel pipe, perforated 306 stainless steel sheet, 10mm mild steel bar, ceramic insulation, ceramic insulation rigidiser, and wood stove gasket tape.
The materials for testing are: beech wood chips (pre-cut and soaked for periods of 0h, 4h, 8h, 12h, and 24h), fire starters, digital kitchen thermometer, digital clock, 20cm stainless steel pot, flame diffuser, lighter, white rice grains, water, lab coat, goggles, heat resistant gloves.

3.2 Construction of the wood stove

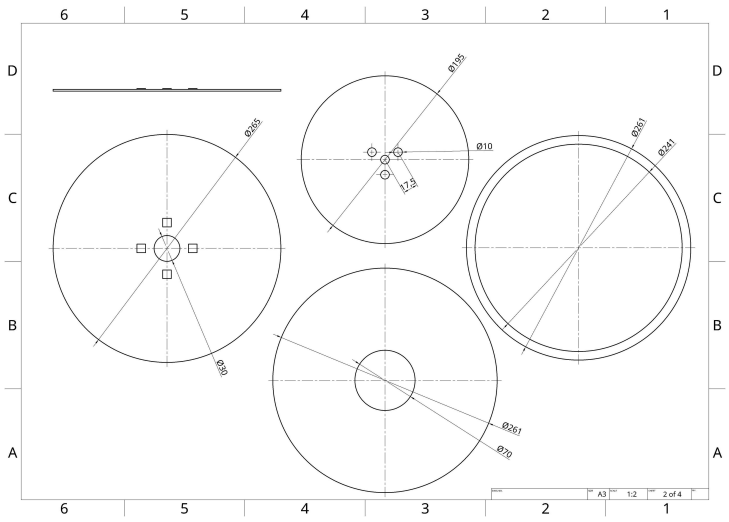


Figure 16. Circumference dimensions

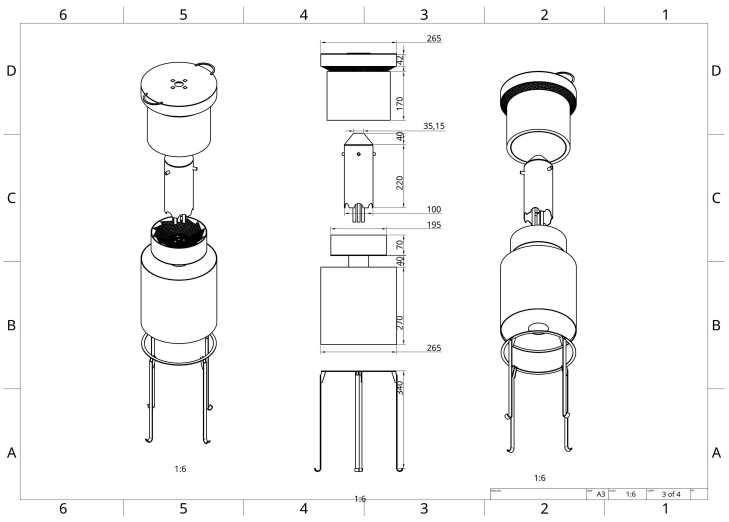


Figure 17. Technical drawings

The stove was made according to the design and dimensions shown above in Figure 16 and 17, with a variety of engineering drawings produced before manufacture, with the help of GF Custom Fabrications. During production, some modifications were made due to both problems and improvements. The original manufacture was made using prefabricated galvanised steel tubes, however, due to them warping from the heat when welding (See Figure 19), the health hazards of welding galvanised steel due to the fumes produced, and the potential health hazards for a user when turning on the device, we instead had to build a custom sheet metal roller so we could roll 2mm mild steel into the correct dimensions (See Figure 19). The body is thus made of rolled sheet steel welded at the seams, with sheet steel also used to make the caps and flat pieces for different components.

Figure 18. Warped galvanised steel

Figure 19. Handmade sheet metal roller

Other changes made during manufacture included adding a lip to the outside of the design, which can be screwed shut to allow for a tighter gasket seal (See Figure 20). The ceramic insulation was also thickened and placed on the bottom of the casing to reduce air entering through gaps between the steel and reduce the risk of burning to users (See Figure 20). The number of holes in the wood funnel top was also reduced, as the airflow through that section did not need to be very high, and this would help prevent the wood from getting stuck in the funnel by producing a smoother surface (See Figure 21). A gasket was added to the lid to ensure gases would exit only through the central funnel and to increase the pressure within the device to improve efficiency (See Figure 20). The last change made was to the stand, which has 3 legs instead of 4 to improve stability on uneven ground (See Figure 20).

Figure 20. Manufactured design

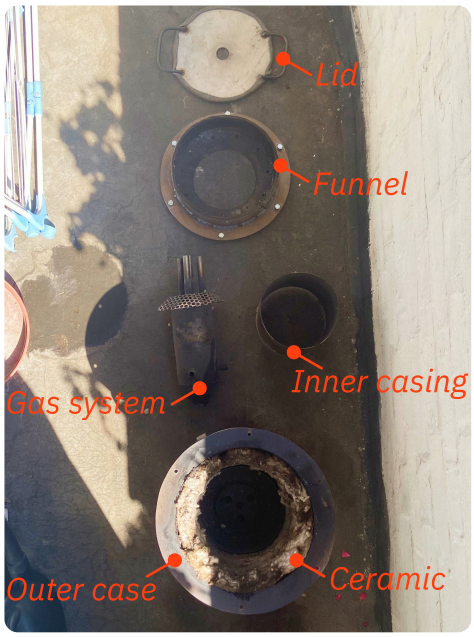


Figure 21. Components laid-out

Figure 22. Inside of design

3.3 Preparation of wood samples

The beech wood chips purchased were already pre-dried in a kiln to an expected humidity of 14% and 19% and sized between 11mm and 31mm in length. The humidity of 0.4kg dry batches was increased by soaking it for periods of 0h (To act as control), 4h, 8h, 12h, and 24h, which is above the minimum required to run the device at 294.5g, and then left to air-dry for 30min to remove any surface water. The different periods correspond to expected water uptake of around 460g/m^2, 558g/m^2, 642g/m^2, and 890g/m^2, respectively (Michalec & Niklasova, 2006), and the weight differences can be used to determine the moisture content.

Figure 23. A wood sample in its bag

Figure 24. Weighing of wood samples

3.4 Experimental procedures

Before each test, the weight of the wood will be measured for each humidity and the moisture content calculated using the following equation (Gute et al., 2022) (where MC = moisture content, M_{wet} = mass of wet wood, M_{dry} = mass of dry wood):

$$MC = \frac{M_{wet} - M_{dry}}{M_{dry}} \times 100$$

Each test will use the same standard operating procedure to light, run and shut down the small-scale wood gasifier, with an expected run time of 2 hours per test:

1. Fill the hopper halfway with wood chips, then light them with the lighter, ensuring the bottom valve is fully open, and the lid is off.
2. When the wood at the bottom of the hopper visibly starts to char and glow, additional wood can be placed up to the bottom of the top wood funnel.
3. We will then wait 15 minutes to ensure the wood at the bottom remains lit, before filling the remainder of the hopper to the top of the lid and sealing the device shut.
4. As the gases build up, the exhaust fumes from the central funnel can be lit periodically until a constant flame is produced.
5. To control the flame's strength, the valve at the bottom can be turned using the additional metal pipe as a handle.
6. Shut down by fully closing the bottom valve and sealing the top funnel with a sheet-metal cap to prevent oxygen from entering the system.
7. Once the device has cooled enough to handle, the ash and remaining wood can be removed.

For each test, once step 4 is completed, a cup of rice (242g) with 2 cups of water will be added to a pot, then placed over the flame with a diffuser below it. At the same time, the timer will begin recording the duration of the flame. The temperature of the rice and water mixture will be recorded with the digital thermometer at the start, then every 5 minutes while the water heats and the rice cooks. When the rice is cooked, the pot will be removed, and the timer will continue running until the flame turns off on its own.

The success of the small-scale wood gasifier at different humidities was evaluated using the maximum water temperature reached, the period during which a temperature above boiling was maintained for sterilisation, and the duration of the flame to assess its ability to support longer cooking times without refilling the hopper. The sterilisation period has been measured to be at least 131 seconds at 60° and 19 seconds at 65° for the most resistant bacterial strains (Spinks et al., 2006). Meanwhile, the successful cooking of rice will be evaluated through an objective taste test.

4. Results and discussion

4.1 Moisture content

The moisture content was calculated using the equation from section “3.4 Experimental Procedures”. For the 0h soaking time, the moisture content was estimated to be between 14% and 19% as the wood chips had been pre-dried in a kiln. The mass of dry wood chips for all soaking times was kept constant at 400g.

Soaking time (hours)	Mass of wet wood (g)	Moisture Content (%)
0	---	14% to 19% (estimated)
4	761	91%
8	808	102%
12	886	122%
24	908	127%

The very rapid absorption of water from the wood is likely to have occurred due to the high surface area-to-volume ratio of the wood chips, resulting from their small size, which meant the moisture content ended up much higher than expected. Due to the small difference of only 5% increase between 12 and 24 hours, compared with 20% between 8h and 12h, the wood likely reached its saturation limit. While the high moisture content could prove to be an issue if I were unable to light the wood at 91% humidity, using fire-starters and a forced draft, I was able to start the gasification process even at that base humidity. Testing would later also be conducted with fresh-cut wooden sticks, which have an estimated humidity of around 60%, to determine whether they run more efficiently than the wood chips.

4.2 Issues with testing

Testing the wood stove proved difficult due to a mix of overlooked factors, design issues, and manufacturing issues. This meant that the testing ultimately differed from the experimental procedures highlighted above, with the successful cooking of rice, sterilisation of water, and the duration of the flame being partially evaluated.

Initial testing began by testing if the start-up procedure worked with the 0h wood chips. While this was successful and produced the first gasifier flame (See Figure 25), the flame lasted only 4 minutes before the wood chips had all burnt out. This, I hypothesised, was due to the small size of the wood chips, which prevented them from retaining heat internally long enough to maintain high enough temperatures for the cracking of hydrocarbons, with them only glowing for a short period of time (See Figure 26). This was later proven through successful testing with larger, fresh-cut sticks.

Figure 25. First gasifier flame

Figure 26. Glowing wood chips

The use of the small wood chips also created a secondary problem that affected efficiency. As they burned, they produced very fine ash that would compact and clog the small holes at the bottom of the central pipe, preventing the syngas from passing through the inner tube to exit. While the seal produced in the lid should help avoid this by creating a high-pressure zone above the gasification layers, which would push the gases through the ash, the manufacturing processes used, especially with everything being handmade, meant that the circles were not perfectly round. The sheet metal had also warped slightly, resulting in large gaps that could not be sealed with the gaskets, allowing syngas and water vapour to escape from beneath the lid (See Figure 27). The lack of airflow also resulted in insufficient fresh air entering the system, meaning the temperature would not remain high enough for cracking to occur. This was proven by the need to use a hair dryer to create a forced draft into the system to raise temperatures when using the wood chips (See Figure 28). Another issue I had was with wind, where the high wind during the testing day caused the flame to be quickly snuffed out as it exited the central pipe (See Figure 29).

Figure 27. Escaping gases

Figure 28. Forced draft

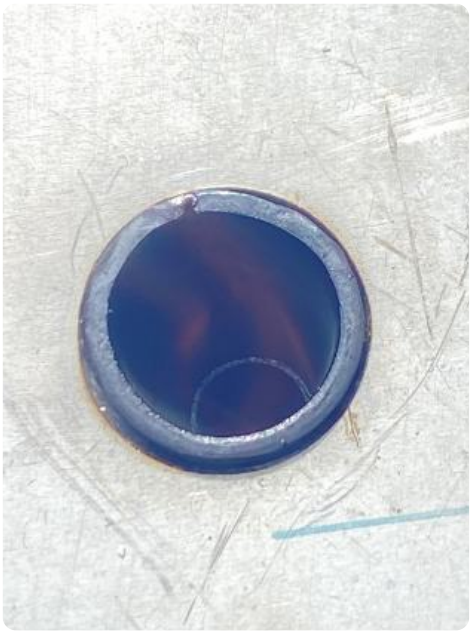


Figure 29. Flame being snuffed

4.3 Success with testing

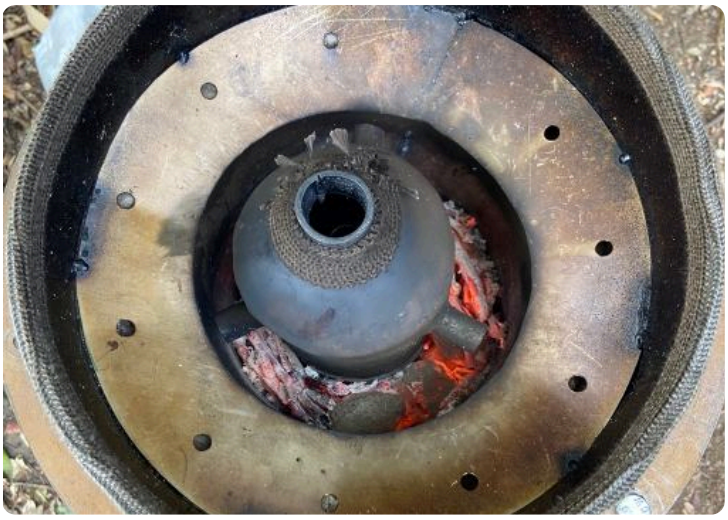


Figure 30. Gasifier with sticks

Figure 31. Cooking rice

However, some successes with the design did occur. Given my hypothesis earlier, and to test the impact of fuel choice on the success of the gasifier, I cut fresh wood sticks of sizes of around 50mm in length, which, when burned (See Figure 30), ran uninterrupted for up to 12 minutes before high winds and compacted ash eventually snuffed it out.

This allowed me to test how quickly it would heat water with rice in it over 8 minutes (See Figure 31). The water and rice in the steel pot had an initial temperature of 12°C and reached 42°C, indicating that the gasifier released 85,014 J over 8 minutes, or 10,627 J/min. This means that, based on the required energy calculated in section “2.4 Design iteration”, it would take 33.3 minutes to boil the water and cook the rice. Their success in running for longer also indicates a more ideal fuel choice than the fuel chips and supports the hypothesis in section “4.2 Issues with testing”. The calculations below show how the 33.3 minute timing was calculated (where Q = energy, m = mass of water, c = specific heat capacity of water, ΔT = temperature change):

$$Q_{water} = m * c * \Delta T = 0.474 \times 4184 \times (42 - 12) = 59,597\ J$$

$$Q_{rice} = m * c * \Delta T = 0.242 \times 1800 \times (42 - 12) = 13,068\ J$$

$$Q_{pot} = m * c * \Delta T = 0.82 \times 502 \times (42 - 12) = 12,349\ J$$

$$Q_{total} = 59,497 + 13,068 + \ 12,349 = 85,014\ J$$

$$\frac{Q}{time} = \frac{85,014}{8} = 10,627\ J/min$$

$$Time = \frac{353,561}{10,627} = 33.3\ min$$

I was also able to measure the rate at which the wood dried when placed on top of the hopper. To do this, I placed the 8h, 12h, and 24h wood around the top of the hopper and recorded how long it took for them to go from physically wet to being able to be lit by a lighter, in 5-minute intervals. Figure 32 shows the progression of the 8h wood chips over 35 minutes, until they were dry enough for ignition. The 12h and 24h wood chips took 50 and 55 minutes, respectively.

Figure 32. Drying of 8h wood chips over 35 minutes

The final measurements I took were of different sections of the design and how hot they got. The lid was the warmest component that a user could touch, reaching a maximum temperature of 84°C. The sides of the casing instead were more protected by the insulation, reaching 46°C. The handles for the lid also warmed up to 38°C, mostly due to gas escaping through the gap between the lid and the side walls, which could be mitigated with a better seal. However, the most successful component was the valve at the bottom, which only reached 26°C. As a component the user would be interacting with repeatedly, this was an ideal result.

5. Redesign and conclusion

5.1 Evaluation of the design

When looking back at our research question “Can a small-scale wood gasifier stove be used for cooking and heating with wet wood post-flood in Vietnam?”, the testing conducted above shows some promise in its ability to fulfil this; however, some issues need to be redesigned and tested before a conclusive answer can be given.

For point 1 under section “1.8 Research question”, the ability of the wood gasifier stove to boil and sterilise water, and then cook rice in that water, was partial but very promising. The stove could gasify wood only for short periods, ranging from intermittent ignition of the gases to a maximum of 12 minutes of continuous flame. From my analysis of the results, this was due to a mix of factors. Firstly, the need for a forced draft to sustain the reaction indicates that not enough air was entering the system to keep the wood hot enough for cracking. This, combined with the compacted ash, meant that gas flow through the system was very limited, so a better way to handle the ash needs to be designed. Finally, a better sealing lid would improve gasification by forcing gases to exit through the middle flute, so that would have to be redesigned as well. The partial success for this point came from the 12 minutes during which the flame was burning and the 8 minutes during which the water and rice were heated, which allowed me to estimate that the water and rice would be cooked in around 33 minutes. Another success in the design was the secondary oxygen pipe, which clearly helped combust the syngas, as shown in Figure 29 in section “4.2 Issues with testing”, where the gases burn inside the pipe.

Point 2, the maximum moisture content of wood that the wood gasifier stove can handle, was inconclusive. The monorator design dried all wood samples featuring an initial moisture content of 127%. Since gasification was not occurring during testing, I cannot evaluate whether too much energy would be lost to keep the temperature high enough for cracking. Improving the design's efficiency through improved gas circulation, ash management, and a better-sealed lid would help evaluate and improve this process.

For point 3, the stove’s portability proved successful. The stove, even with thicker steel than originally planned, only ended up weighing 14.3kg empty, so it was easy to lift and carry around. However, something was missing in the original CAD design: handles for the main body, which proved crucial, as the lips added to the manufactured design were for carrying the device around. The device's dimensions were also within the required limits, occupying 369mm by 310mm by 310mm of space, within the 450mm cube I had constrained it to.

Point 4, the stove's ability to warm the surrounding area to warm victims, was partially successful. While the flame generated heat, no measurements were taken beyond the casing temperature, which could be used to evaluate how well the stove heated its surroundings or someone standing next to it. Further evaluation of this should be conducted in later testing.

Finally, for point 5, the design partially meets the Moving Energy Initiative’s 11 design considerations highlighted in section “1.5 Review of existing solutions”. The design considerations that haven’t already been discussed above include its ability to work with existing utensils. This was successful, as pots could be placed next to the flame; however, for a redesign, I would like the pots to be able to go directly above the flame. Its stability was proven to be very strong, with no issues throughout the day of testing, even on uneven ground. The device also proved very durable, although some damage occurred to the ceramic fibre insulation during cleaning, so a redesign to better protect it is

warranted. Safety was also successful, as the valve below controlled airflow easily and effectively, without heating to dangerous levels. The outer casing was not hot enough to immediately burn someone.

A final aspect raised when showing the device to an external validator, who raised the point that ‘the inability of users to see the wood as it was burning would create distrust in the stove’ (Information provided during a personal interview, May 15th, 2026. The interviewee's name has been withheld by request to protect their anonymity). This is an issue I had not considered, as users may become uncertain whether the wood inside the device is actually burning, leading to distrust of the product. Potential redesigns for this could include a simple viewing window or port in the lid, although that would greatly complicate manufacturing, so implementation should be assessed carefully. When testing the device's assembly, the external validator also struggled to align all the components so everything would sit straight, with major issues with the 4 small pipes in the inner tube, the different middle-sized tubes, and the orientation of the 6 screws for the top gasket. While this can be mostly attributed to imperfect manufacturing, the design should be improved to facilitate cleaning and disassembly of the device.

5.2 Redesign

The overall design has slightly changed from the original and manufactured iterations. The redesign does not contain the 4 air pipes from the first design but instead will be fed as a form of updraft gasifier, with all the oxygen entering below the ash layer, passing through the various stages, and eventually leaving through the single pipe (See Figure 33). The redesign also removes the internal components, such as the inner pipe, hopper, and ash funnel. This both reduces manufacturing complexity and reduces the number of parts, meaning that a user will not struggle to align all the components when reassembling or attempting to clean the device (See Figure 33).

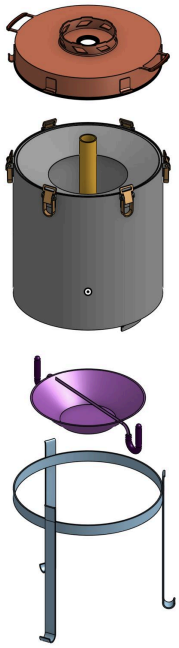


Figure 33. Exploded view

The first issue to be fixed was the management of the ash and its effects on gas flow. The main issue with the ash was that it compacted at the bottom of the grate, severely blocking the small holes and gas intakes of the inner pipe, which in turn reduced the amount of air entering the system through the gas pipes. To fix this, two main changes were made to the design. Firstly, the size of the holes at the bottom of the grate was increased from 5mm to 20mm to allow more ash through (See Figure 34). Further testing should be conducted across a different range of sizes, from 10mm to 60mm, to determine what works best. The intake holes were also changed from being only at the bottom of the gasifying zone (See Figure 14) to running up the sides, so the finer ash cannot clog them, as it will only be produced towards the bottom of the gasifier.

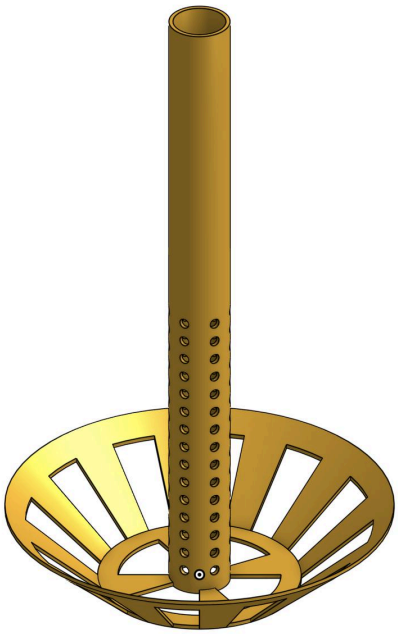


Figure 34. New inner gas funnel

The lid’s sealing has also been improved by introducing FFKM gaskets and O-rings along the edges of the updated lid (See Figure 35 and Figure 36), which can withstand temperatures up to 330°C (Trp Polymer Solutions, n.d.), far above the lid’s maximum recorded temperature of 86°C. The introduction of 6 toggle clamps around the

outside of the casing and lid allows the user to apply pressure to the FFKM gasket (See Figure 33), ensuring a thorough seal at those edges to minimise the chance of gas escaping. These would be easier to align than the screws that were found on the manufactured model (See Figure 20) and would make it easier for a user to place or remove the lid. This also allows a user to lift the whole body through the lid's handles, improving portability.

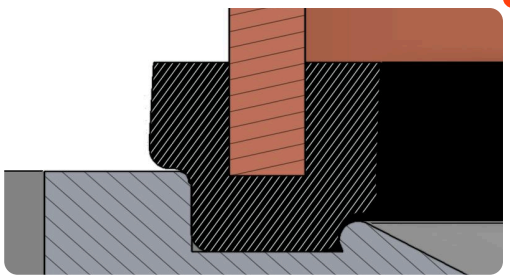


Figure 35. FFKM Gasket

Another major issue I needed to tackle was the lack of wind resistance. While this was overlooked in the original design, the updated design includes a raised windbreaker (See Figure 37) that blocks the wind that would blow away the escaping gas, protecting the flame while allowing a user to place a pot directly above it to cook. This both increased the ease of lighting for a user, helping to meet one of the MEI's 11 needs, and should increase the time a flame lasts. The holes found help feed oxygen to the secondary burn, improving HAP as more of the dangerous pollutants are burned away, and allow me to remove the secondary air pipe from the original design (See Figure 14), reducing complexity.

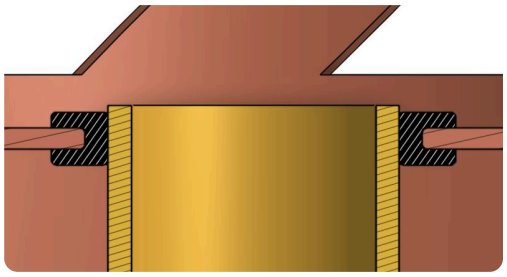


Figure 36. FFKM O-ring

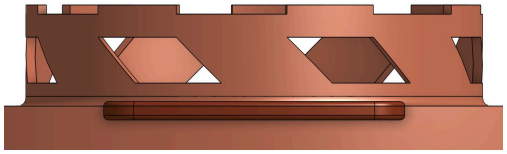


Figure 37. Wind breaker

The most radical redesign, though, was the ash-catcher and air-management system. Throughout testing, I encountered an issue: ash could not be removed without removing all the components inside the device. This was one of the main reasons I was unable to sustain the reaction: the ash clogged the system too quickly, so I needed the bottom of the casing to be removable. While I considered adding a door that could be opened and closed from the side, the easiest solution I came to was to have a bowl below the opening, connected to a tiered ratchet system (See Figure 38). This allowed a user to twist the bowl up or down the ratchet using the external handles, controlling the airflow entering the system. If it fills with ash, it can be lifted off, disposed of, and then placed back quickly and easily. The other advantage of this system is that it allows a user to light the system from below, improving ease of use and trust, as users see whether the embers are still lit without removing the lid.

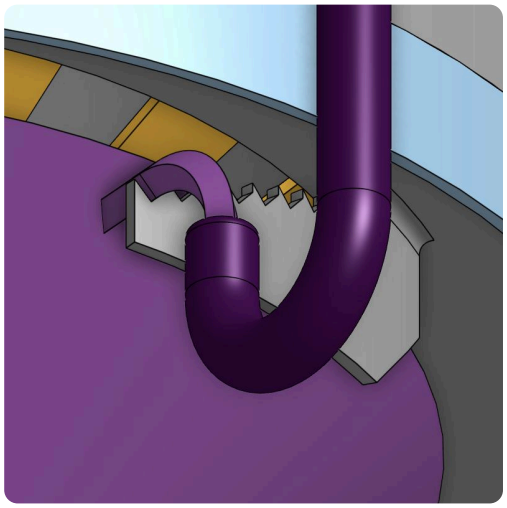


Figure 38. Ratchet system

Other minor tweaks included incorporating the side walls of the hopper into the top of the lid to improve the seal produced by the lid, adding silicone to the handles to improve the heat resistance so a user could interact with them without getting burnt, and remodelling the stand to be 3 legged to improve stability, and easier to manufacture (See Figure 39). The overall changes also reduce the device's weight to a calculated 11.4kg and decrease the number of parts before manufacture from 72 to 24.

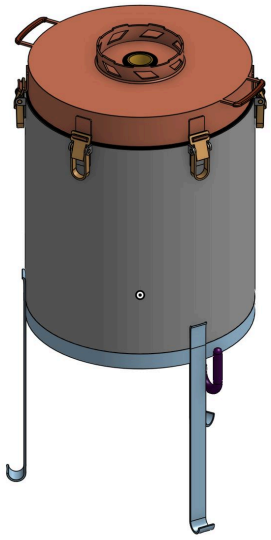


Figure 39. Whole body

5.3 User manual

With a good understanding of how to use the device and an updated design, I was able to create a user manual in both English and Vietnamese (See Figure 40 and Figure 41).

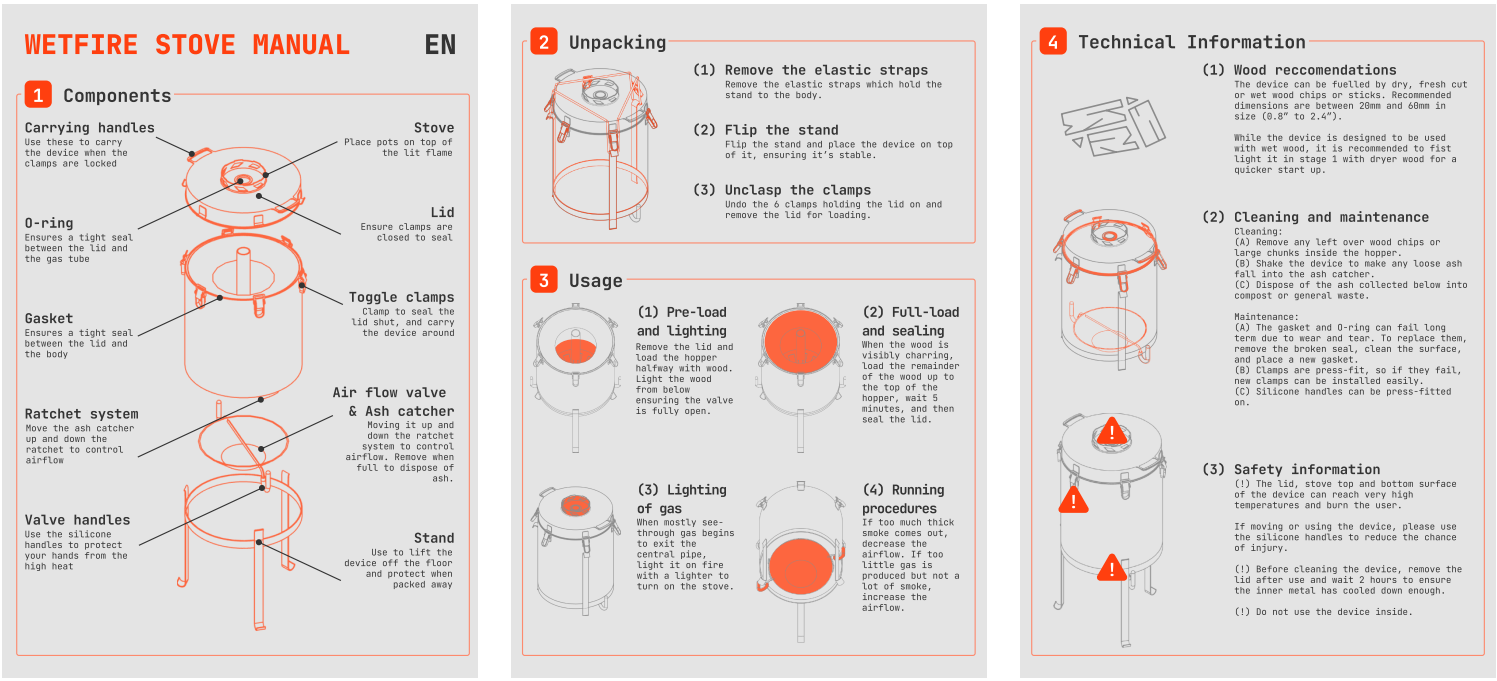


Figure 40. English user manual

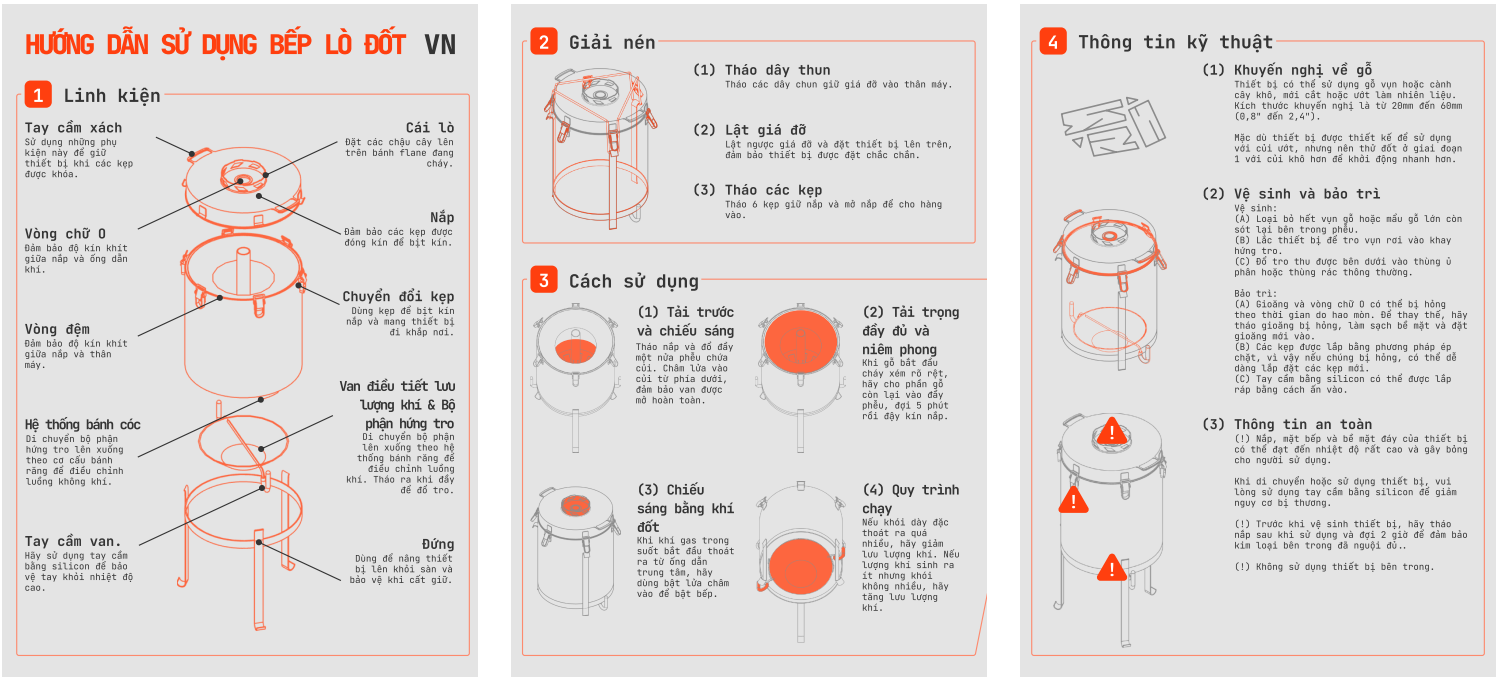


Figure 41. Vietnamese user manual

5.4 Updated user journey map

The following updated user journey map (See Figure 42) compares how users used to have to react to and live with a flood, and how, with my proposed intervention, they could alleviate some of the stresses they had to deal with.

The distribution of the stove could occur from both institutional or local aid, with neighbours helping neighbours. When looking at the timeline, since the stove is intended for permanent installation in a home, it would ideally be delivered before the flood. However, due to its size and weight, it is still possible to deliver it in an emergency after the flood.

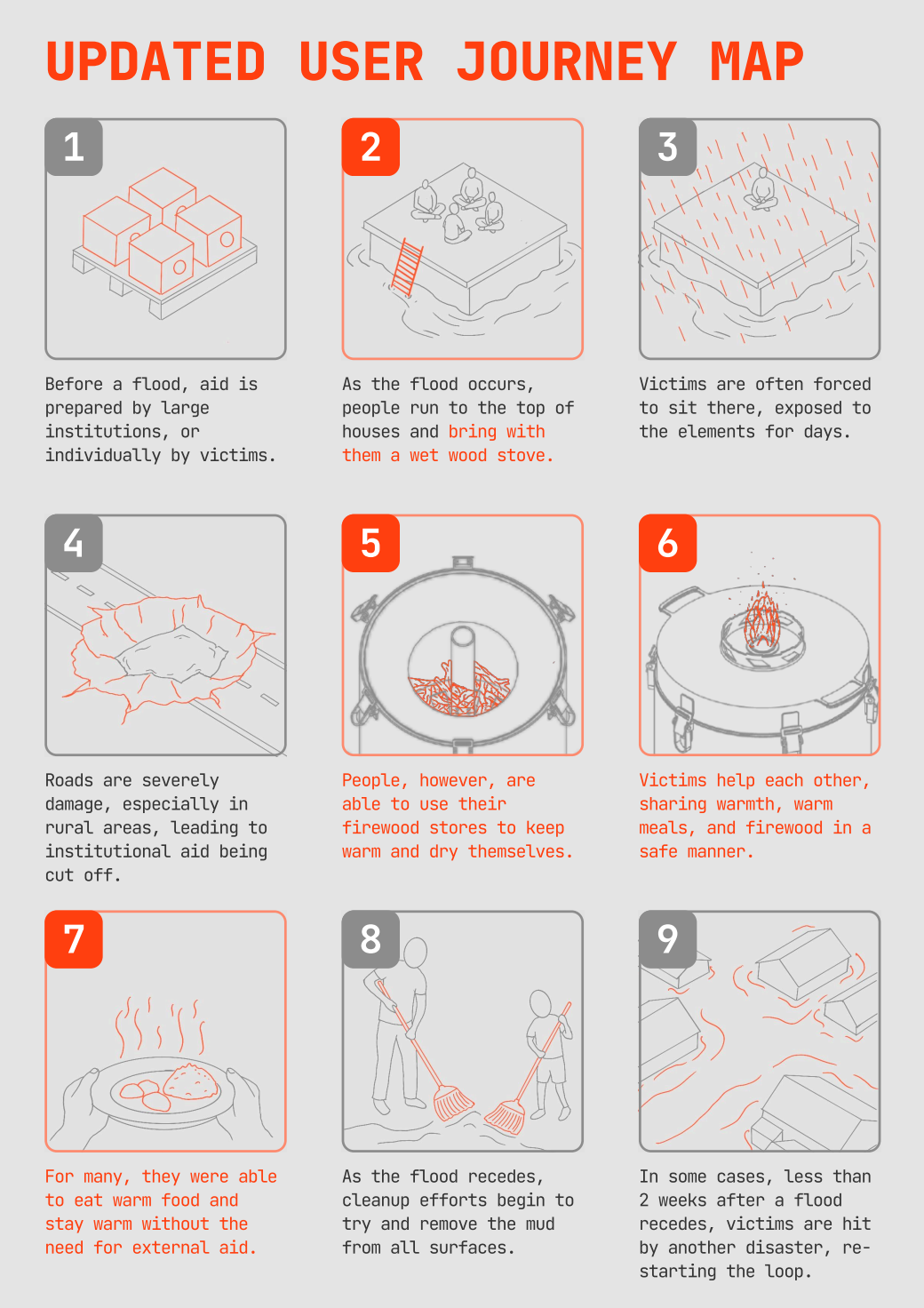


Figure 42. Updated user journey map

5.5 Project management

Gantt charts were used throughout the project in conjunction with the Early-Stage Gateway (ESG), Late-Stage Gateway (LSG) and the final deadline to keep track of work and have specific aims to reach each term:

- ESG: The first term was focused on background research to understand the problem space and potential issues my intervention could tackle. This section was divided into exploring the humanitarian impact, the institutional and local responses to it, and the issues with them. This allowed me to gain various insights that guided me towards cooking post-flood as an issue I could address with an intervention. The following Gantt chart was created at the end of the ESG to plan and keep track of my work during the following terms:

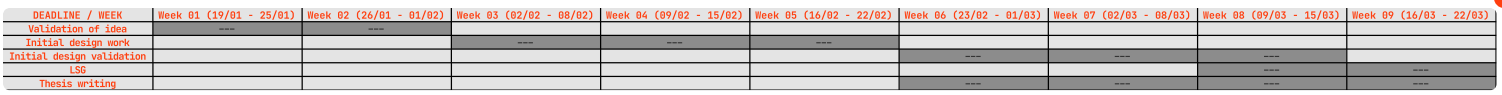


Figure 43. ESG Gantt chart

- LSG: The second term leading up to the LSG focused first on researching solutions that addressed cooking post-flood and cooking with wet wood. I then moved on to validating the idea. I began design work, continuously collecting feedback from both experts on the “Drive on Wood!” forum and other validators on potential designs and issues that could arise. While working on the design, I began the lengthy process of creating and obtaining approval for a Risk Assessment (RA) for testing the stove. The following Gantt chart was created at the end of the LSG to plan and keep track of my work during the term:

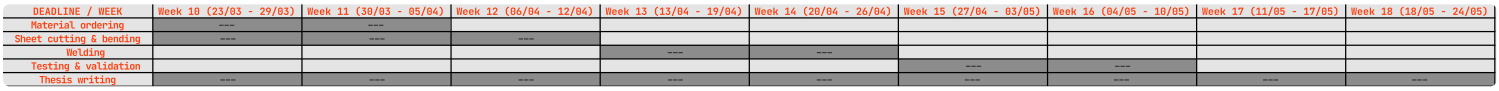


Figure 44. LSG Gantt chart

- Final deadline: Work after the LSG started by going to GF Custom Fabrications to construct the stove, as the university had inadequate workshops to manufacture it there. After it was built, I began structuring and writing my thesis and planning experimental procedures to test it. Testing finally occurred 4 weeks later than originally planned due to issues with the RA approval. Once testing concluded, I was able to collect and analyse the results and finish writing my thesis, staying on track for the first and final drafts. Figure 45 shows the respective Gantt chart.

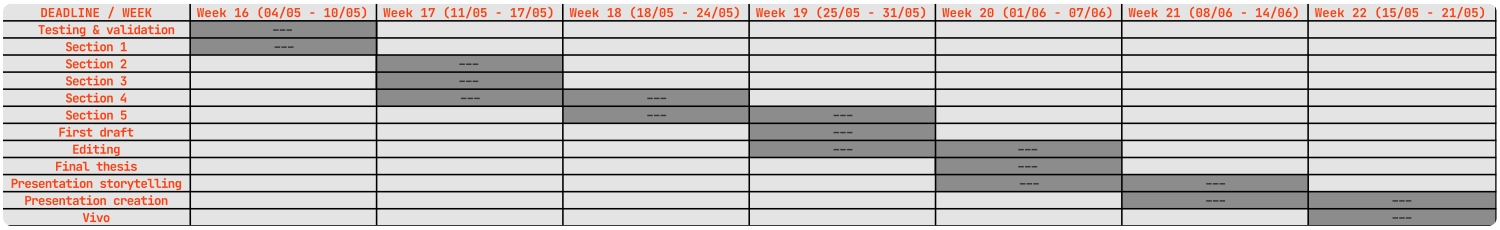


Figure 45. Final Gantt chart

I faced many issues that affected how on track I was with the project. The main one that hindered me was manufacturing, as, while discussing with the workshop technicians at both Ace-X and Hackspace, I quickly found that I would have been unable to manufacture the product on campus due to a lack of equipment and high risk. Fortunately, I found a workshop in Devon that hosted me, where I was able to construct prototypes and the stove over the course of a week. Another massive issue I had was around the RA for testing: A large amount of preparation for both safety equipment and validation of the design had to go into preparing for the testing of the stove, with the university telling me that I would be unable to test it on university grounds due to the fire risk it posed. As such, I had to find alternative private land to test it on before my RA would be approved. I was further hindered by the unavailability of carbon monoxide sniffers on campus, which added another 2 weeks to my wait time before I could commence testing.

Stakeholders were managed across different sources. The main ones I worked with were found on the “Drive on Wood!” forum, whose expertise helped shape the design of the stove and gasifying system. Stakeholders from the UN, UNICEF, and the VDDMA were contacted to gain additional insights; however, success was limited, with greater success found through contacts with various NGOs working on stoves and humanitarian aid.

5.6 Future scope of the project

Future work on this project should focus on 3 key things. First, the manufacture and testing of the updated design should be conducted to determine its viability and usability, and to identify any other changes that should be made. This would also allow testing of the device's ability to sterilise water, cook food, heat the surroundings for warmth, and determine the maximum moisture content of wood the device can handle. Second, the size of the wood chips should be tested in the updated design, ranging from 10mm to 60mm, to determine what works best for the device and whether any redesigns to accommodate smaller or larger pieces should be conducted. The final thing that should be tested is the HAP exposure, specifically the PM2.5 and CO produced, and if there are any ways to improve it, such as by cold filtering the gas (where one cools down the gas before burning it). Together with this testing, deployment of the device should also be explored, not just in Viet Nam but also in other flood-prone areas whose populations rely heavily on firewood.

5.7 Conclusion

This project was an ambitious attempt to produce a stove that addresses an issue affecting a multitude of people, not just in Vietnam but worldwide. While the prototyped design was only partially successful, the process of research, design, build, and test gave me a great deal of knowledge and experience in creating and improving the design of small-scale gasifier stoves. Future testing of the updated design should build on the promising results from the tests I conducted in this paper. Additional resources should be invested in this project, all to help the millions of people worldwide who use biomass as their primary fuel and are unfortunately hindered by the limitations and health risks of using wood after a flood.

6. References

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